

Food Waste Management Strategies of Selected
Restaurants in Benguet

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Bachelor of Science in Hospitality Management

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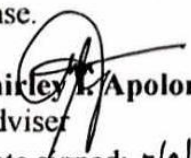
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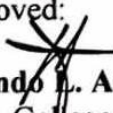

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ABSTRACT

This qualitative study investigated the strategies employed by restaurants to manage and reduce food wastes. Utilizing purposive sampling, a total of 20 restaurants, owners and managers were selected as key informants. Data were gathered through interviews to explore their practices from food preparation to waste disposal. Then analyzed through thematic analysis.

This study revealed the food waste management strategies, challenges and support. Basing from the results restaurant implement four strategies. First is portion control. It is a systematic monitoring of inventory and calibrated portion sizing to minimize overproduction. Another strategy is repurposing, which converts food scraps into broths, fillings, sauces, or new dishes. The third strategy is composting, which involves organic waste segregation for composting or use in bio-waste recycling. Another strategy is having staff training, engaging employees in best-practice waste handling through regular education programs.

There are also challenges which the restaurants faced in implementing the strategies. These are cross contamination, mixing old and new stocks, incorrect measurements, lack of time, and improper waste segregation.

Challenges may affect their food waste management strategies but they were able to established support which helps in improvement of managing their food waste. These supports are reliable and consistent supplier, bench marking, and outside training sessions.

Keywords: *Food, Waste Management, Restaurants, Waste Reduction Strategies, Waste Management Challenges, Support*

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CHAPTER I

INTRODUCTION

Background of the Study

Food waste is a critical global issue that affects environmental sustainability, economic efficiency, and social equity. Globally, the Food and Agriculture Organization of the United Nations estimates 1.3 billion tons of food are wasted annually, equivalent to one-third of all food produced for human consumption (FAO, 2021). The restaurant sector is a significant contributor to this waste, particularly due to overproduction, inefficient inventory management, and consumer behavior.

At the international level, food waste management practices have increasingly emphasized reducing the environmental footprint of the food service sector. According to the World Resources Institute (2022), food waste costs the global economy over \$1 trillion annually, with a significant portion originating from the hospitality industry. Restaurants often incur losses due to spoilage, oversized portions, and food safety regulations that mandate the disposal of unsold food. To address this, innovative programs such as Too Good to Go Which operates in Europe and the discounted prices, effectively reducing waste while generating additional revenue. Furthermore, inefficient supply chains and inaccurate demand forecasting contribute to over-ordering and inventory excess, resulting in further spoilage. Technological innovations like Fresho's AI-powered platform have emerged as effective solutions, streamlined the ordering process and helped restaurants and suppliers minimize overproduction and food waste (The Australian, 2024). In addition, consumer expectations, particularly in countries like the United States, significantly influence food waste. As noted by

the National Academies of Sciences, Engineering, and Medicine (2020), cultural norms around value-for-money often encourage larger portion sizes, which leads to higher levels of plate waste. However, research shows that offering smaller, "right-sized" portions can reduce waste without compromising customer satisfaction.

Food waste presents significant challenges in the Philippines, impacting economic, environmental, and social dimensions. In 2021, the Philippine Institute for Development Studies estimated that households alone generated approximately 9.4 million tons of food waste, with commercial establishments such as restaurants and hotels contributing further to the problem. Despite widespread hunger in the country, where over 13.5 million Filipinos experienced involuntary hunger at least once in 2022 (Social Weather Stations, 2022) food service businesses continue to discard large quantities of edible food. According to WWF-Philippines (2021), much of this waste is avoidable, often resulting from over-preparation and inaccurate demand forecasting. To address this issue, WWF-Philippines launched "The Sustainable Diner" initiative, which engages government agencies, food service businesses, and consumers in promoting sustainable dining practices. The project emphasizes the use of seasonal ingredients, plant-based menu options, and proper waste segregation to minimize food wastage in the food service sector. However, a 2020 study by WWF-Philippines revealed that many food service workers remain unaware of the amount of food they waste daily and the impact this has on both business operations and the environment. This lack of awareness contributes to inefficient food handling, poor inventory management, and the unnecessary disposal of still-edible food.

Based on the Physical and Socio-Economic Profile (2016) of La Trinidad, this municipality is one of the Central Business District (CBD) in Benguet, located in the Cordillera Administrative Region. La Trinidad is considered the Central Business District in Benguet due to its strategic location as a gateway to the Cordillera region and its role as a center for education, banking, finance, and socio-economic activities. It's also a major trading and marketing hub for the province. This is the reason why it encounters growing concerns regarding food waste, particularly from its restaurant sector.

The lack of efficient food waste management systems in local food establishments exacerbates the problem. Common practices such as overproduction, inadequate inventory management, and improper food handling are frequently observed, especially among small to medium-sized restaurants. According to a 2021 report by WWF-Philippines, many food service workers remain unaware of the volume of food they waste and the environmental consequences of such practices. Improper disposal methods, including the dumping of used cooking oil into drainage systems, have further prompted proposals for ordinances banning these harmful practices (Philippine News Agency, 2018), underscoring the urgent need for improved waste segregation and responsible disposal in the food sector. Moreover, local government units are actively seeking sustainable waste management technologies, as the Alno dumpsite, the town's primary waste facility, is nearing full capacity (Herald Express, 2023). Although segregation at source is mandated by local policies, compliance remains inconsistent among commercial establishments, including restaurants. On a positive note, community-based initiatives such as the Green Market program and the Bantay-Basura volunteer monitoring project are making

progress in promoting waste segregation and reduction. However, the success of these programs depends heavily on the active participation of the restaurant industry. Establishments in La Trinidad could greatly benefit from implementing practices such as seasonal menu planning, portion control, repurposing leftovers, donating surplus food, and adopting food waste tracking technologies strategies that have demonstrated effectiveness in similar settings. Furthermore, the findings of this research can help restaurant owners improve operational efficiency, guide local policymakers in crafting environmentally responsible policies, and contribute to broader sustainability goals within the community.

This study will benefit several stakeholders who play a role in food waste management and its sustainability within La Trinidad, Benguet. Primarily, restaurant owners and managers will gain meaningful insights into practical and effective food waste management strategies that can be implemented within their daily operations. These insights may help them optimize food handling processes, minimize waste, reduce operational costs, and improve overall efficiency ultimately contributing to both environmental responsibility and business profitability. It also benefits the customers as improved practices of the restaurant can lead to a better food quality, good portion sizes of the food and more dependable dining experiences. Additionally, the findings of the study can support Local Government Units (LGUs) and environmental agencies by providing localized data and evidence-based recommendations that can inform the creation or enhancement of policies, regulations, and programs aimed at reducing food waste within the food service sector. Furthermore, the study will be valuable to future researchers and students, particularly those studying hospitality management, environmental science, public administration, and sustainability. It

can serve as a useful reference for academic work, policy analysis, and further studies focused on food systems and waste management in similar communities. By filling the current research gap in the local context, this study also contributes to the broader knowledge base on sustainable food practices. Lastly, the general public and the local environment are indirect but important beneficiaries. Improved food waste management by restaurants can lead to less strain on local waste disposal systems, reduced greenhouse gas emissions, and a healthier, cleaner community. In the long term, promoting responsible food waste practices supports environmental conservation, community health, and sustainable development in La Trinidad.

This study aims to explore the current food waste management strategies employed by selected restaurants in La Trinidad, Benguet. Specifically, it seeks to identify the types of practices restaurants are currently implementing, how these practices are carried out, and the challenges that hinder their effective application. The findings of this study are expected to provide valuable insights into the food waste landscape in La Trinidad and inform recommendations for more sustainable, cost-effective, and practical food waste management solutions tailored to local conditions.

Theoretical Framework

This study is grounded in the Theory of Planned Behavior (TPB) and is supported by the Food Waste Hierarchy (FWH) and the concept of Sustainable Restaurant Practices. These theoretical frameworks collectively provide a comprehensive lens for examining the behaviors, strategies, and challenges that restaurants in La Trinidad, Benguet face in managing food waste.

The Theory of Planned Behavior (TPB) as proposed by Ajzen (1991) provides a behavioral foundation for this study by explaining the psychological and social factors that influence the decisions of restaurant owners and staff regarding food waste management. In this theory, individuals' behaviors are shaped by their attitudes toward the behavior, subjective norms (such as industry expectations or community standards), and perceived behavioral control (the extent to which individuals feel capable of performing the behavior given available resources, time, and knowledge). This framework enables the study to explore not only what food waste management strategies are employed but also the underlying reasons and motivations behind the adoption or non-adoption of these practices.

The Food Waste Hierarchy (FWH), as outlined by Papargyropoulou et al. (2021), serves as a guiding structure for analyzing and categorizing the various food waste management strategies implemented by restaurants. It prioritizes actions from the most to the least environmentally preferred: waste prevention, reuse (e.g., donation), recycling (e.g., composting), and disposal. This hierarchy helps the study assess where the participating restaurants fall in terms of sustainable waste management and how closely their practices align with global best practices for minimizing food waste and environmental harm.

The concept of Sustainable Restaurant Practices (Singh & Ghosh, 2023) complements these frameworks by offering a practical perspective on how food waste reduction is integrated into daily restaurant operations. It focuses on environmentally and economically sustainable actions such as inventory control, portion sizing, staff training, and responsible sourcing. This concept helps the research explore how operational practices contribute to or hinder effective food

waste management, particularly in the context of small to medium-sized restaurants operating within the resource-constrained environment of La Trinidad's Central Business District.

Food waste has become a significant concern worldwide, not only for its environmental impact but also for its economic and social implications. Due to its advanced development, La Trinidad is considered one of the Central Business Districts of Benguet. The municipality is relatively the most developed settlement in the province of Benguet because of its proximity to the city of Baguio. It is the gateway of the southern lowlands, into the Cordillera Region. Despite this, there is a lack of localized research that explores how restaurants in this area manage their food waste, what specific strategies are being implemented, and the challenges they encounter. Studying this issue is essential in identifying current practices, raising awareness about sustainable food management, and supporting the development of more effective waste reduction strategies.

Statement of the Problem

The purpose of this study is to find out the strategies of food waste management in selected restaurants in La Trinidad, Benguet. This study sought to answer the following questions:

1. What are the food waste management strategies employed by the restaurants in selected restaurants in La Trinidad, Benguet?
2. What challenges do restaurants face in managing food waste?
3. What kind of support would help improve food waste management?

Chapter II

METHODOLOGY

This chapter presents the research design, population of the study, data gathering tools, data gathering procedures, and data analysis.

Research Design

The study utilizes a qualitative approach to explore the food waste management strategies implemented by selected restaurants in La Trinidad, Benguet. Qualitative research is particularly suited for this investigation as it allows for an in-depth understanding of the practices, motivations, and challenges faced by restaurant operators in managing food waste. This approach is essential for capturing the nuanced perspectives of individuals directly involved in food service operations, providing insights that quantitative methods may overlook (Carter et al., 2021).

Population and Locale of the Study

This study was conducted in selected restaurants within La Trinidad, Benguet, focusing on key commercial areas such as Longlong and the area from Km. 3 to Km. 6. These locations represent some of the most active food service zones in the municipality, housing a wide variety of restaurants that cater to both locals and visitors. The area is characterized by its growing urbanization and commercial activity, making it an ideal setting for investigating food waste management practices in the restaurant industry. There are 20 respondents, composed of restaurant owners and managers operating within the identified locale. The type of restaurants that the researchers interviewed include family-

style restaurants, restaurants that serve large portions of food for sharing. Another type is a casual dining restaurant, a restaurant that strikes a balance between fine dining and fast food. Lastly, Bistro is a small and casual restaurant that serves simple meals in modest settings. These types of restaurants led the researchers to explore the food waste management strategies of the restaurants.

The selection of restaurants followed specific criteria to ensure the relevance and quality of data collected. These criteria include excellent customer service, longevity or consistency in the food service industry, strategic location and accessibility, availability of combo meals on the menu, affordable price, and amount of portion sizes. Additionally, only restaurants with at least a base on Google Reviews were considered, ensuring a baseline of customer satisfaction and operational standards.

Restaurant owners and managers are directly responsible for overseeing the daily operations of their food establishments, and their insights are essential for understanding the strategies implemented in managing food waste, how these strategies are applied, and the challenges encountered during the process. The researchers use convenience sampling, selecting restaurants that are accessible and whose owners or managers are willing to participate in the interview process. The researchers need to gather meaningful data while accommodating logistical and time constraints.

Data Gathering Tools

The researchers conducted interviews with selected respondents, specifically restaurant owners and managers, to gain deeper insights into their food waste management practices. The interview consisted of several sections,

including demographic information (name of the restaurant), the strategies employed by the restaurants, challenges faced by the restaurants in managing food waste, and support that would improve food waste management of the Restaurant. Through these interviews, the researchers obtained deeper insights into the food waste management strategies employed by restaurants and the key factors affecting their effectiveness.

Data Gathering Procedures

The research process began with the checking and approval of the study title by the instructor, followed by a research orientation conducted by the research office as a guide through the research process. The researchers proceeded with the creation of a proposal matrix, which was later presented for evaluation and feedback. Afterward, Chapters 1 and 2 were crafted, which were reviewed by the research director, leading to a revision of the paper. This was then followed by the selection of a research adviser while simultaneously awaiting approval from the Research Ethics Committee (REC). Once approved, validation of the interview questions was conducted to ensure credibility and clarity. Afterwards, an endorsement paper was attached for the proposal defense. The next phase involved conducting interviews with selected respondents to gather quality data, which was then presented and analyzed using thematic analysis.

Data Analysis

This research was guided by Braun and Clarke's (2021) Thematic Analysis, which is particularly effective in exploring experiences, behaviors, and motivations. This aligns with the study's goal of exploring the food waste

management practices from the perspectives of selected restaurant owners or managers in La Trinidad, Benguet. Thematic Analysis allows for the systematic identification of patterns, categories, and themes that emerge from participants' responses. It helps to uncover the various strategies employed by restaurants in managing food waste, how these strategies are applied in daily operations, and the barriers they face in the process. By carefully coding, sorting, and interpreting qualitative data, this method provides a comprehensive understanding of food waste management practices in the local restaurant industry, capturing the nuances of how these practices are implemented and the challenges involved.

Ethical Considerations

Before starting the study, the researchers made sure to follow ethical steps. The researchers went to the municipal hall of La Trinidad, Benguet, to ask permission to conduct research and then gathered the information of the registered restaurants. The researchers gave a letter to the selected respondents or restaurants to have a consent form from them and to explain the purpose of the study and how the data would be used for academic purposes only. They were also informed that their answers would be kept confidential.

CHAPTER III

RESULTS AND DISCUSSION

This chapter presents the results, discussion, and interpretation of the data gathered.

First, the researchers gathered information about the strategies or practices they are using in the reduction of their food waste. All the respondents of this study claimed that they are using the *First In, First Out (FIFO)* inventory management strategy. As defined by McMahon (2024), First In, First Out (FIFO) is an inventory management method that ensures the oldest inventory (first in) is used or sold before the newer stock (last in). This method is particularly popular and effective in the restaurant industry due to its focus on managing perishable goods. As defined by Allen (2024) perishable goods are defined as goods that have a fixed or specified shelf-life or expiry date after which they become unsuitable for consumption or use. Additionally, some perishable goods need specialized storage as they're subject to deterioration, spoilage or impairment over a period of time, if not chilled, refrigerated or frozen.

All the restaurants use *First in First Out* because it helps to deliver fresh food, prevents spoilage, supports food safety, and saves money. Since food items have limited shelf lives, implementing *First in First out* ensures that older stock is used before newer stock, thereby minimizing the risk of spoilage and unnecessary waste. Implementing *First in First Out* involves organizing food supplies in such a way that the items received first are placed at the front or top of storage areas, while newer items are stored behind or beneath them. Respondent 1 stated, “*We make sure that we used older stock before using other new stock.*” While respondent 2 added “*To avoid spoilage of the food and to maintain the freshness*

of the food, they cook first the first that they purchase before the latest.”

Respondent 3 also states that “*we redesigned our storage layout so that older ingredients are always more accessible than new ones*”. This system encourages staff to consistently use the oldest stock first, helping maintain product freshness and safety. First In First Out also supports food safety and quality assurance by preventing expired or deteriorating products from accidentally being used in food preparation. As food costs continue to rise and sustainability becomes a higher priority in the hospitality industry, adhering to First in First Out procedures help restaurants control expenses while reducing their environmental footprint. Moreover, the First in First Out system promotes accountability and organization within kitchen operations, ensuring that inventory is properly monitored and utilized, which is particularly important in high-volume settings where inventory turnover is frequent. This practice was supported by Food Waste Hierarchy which prioritizes waste prevention as the most preferred strategy. According to Sembiring (2019) the First in First Out method is interpreted as a method of valuing First in First Out inventory assuming that the first purchased item is the first item used or sold, regardless of the actual physical flow. This was supported by Ragusa et al., (2022) states that the First In, First-Out (FIFO) method is a widely adopted inventory management strategy in restaurant operations aimed at minimizing food waste and ensuring food safety. FIFO involves organizing inventory so that older stock is used before newer deliveries, which helps prevent spoilage and supports efficient stock rotation.

The following are examples of fast-food restaurants who are implementing *FIFO*. First, McDonald's applies the First In, First Out (FIFO) method in managing its inventory to maintain product freshness and quality.

Ingredients such as lettuce, tomatoes, and buns are stocked in a way that ensures the oldest items are used first. Staff are trained to rotate inventory by placing newly delivered items behind the older ones. This practice helps reduce spoilage and contributes to maintaining high food safety standards. By consistently following FIFO, McDonald's ensures that customers receive meals made with fresh ingredients (McDonald's Philippines, 2021). Jollibee implements the First In, First Out (FIFO) system to manage its fast-moving inventory, including chicken, burger patties, and sauces. Kitchen staff are trained to label ingredients with their delivery dates and rotate them accordingly. This ensures that older stock is used before newer ones. Second, by applying First in First Out, Jollibee effectively reduces food waste and maintains the freshness of its ingredients. This practice supports consistent product quality and efficient kitchen operations (Jollibee Foods Corporation, 2020). Third is Kentucky Fried Chicken (KFC) they strictly monitor raw chicken and other perishable ingredients using the First In, First Out (FIFO) system. The restaurant uses color-coded labels and daily inventory logs to track and rotate stock effectively. This process ensures that older items are used before newer ones. By applying First in First Out, Kentucky Fried Chicken (KFC) prevents inventory loss due to spoilage and maintains the freshness of its products. This method also supports consistent food quality and safety standards (KFC Philippines, 2021).

Another strategy that restaurant owners are using is *composting* of 20 respondents, 13 said that they are using this strategy. As defined by the Environmental Protection Agency (2023), composting is the biological process of breaking down organic waste materials, such as food scraps, yard waste, and biodegradable items, into a nutrient-rich soil amendment called compost. This

process involves the natural action of microorganisms under controlled conditions and is widely used to reduce landfill waste and support environmental sustainability.

Some restaurants use composting to reduce food waste, lower disposal costs, and support environmental sustainability. It helps them manage organic waste responsibly. Some of them separate the non-compostable waste and compostable waste to avoid mixing. Respondents stated “*Tuwing oras ng pagsasara, nagsasagawa kami ng pagsusuri sa compost bin upang matiyak na ang tamang mga bagay lamang ang itinatapon dito.*” Some of the restaurants also use organic materials as fertilizers in their vegetable garden to grow plants and crops that are later used in their products. They compost food scraps and other biodegradable waste to recycle nutrients back into the soil. In cases where they do not compost onsite, they store the compostable waste, which is then collected by someone they often know, a farmer or gardener for composting purposes. A respondent said “*Itited mi dagijy arasaw kadagijy ag dawdawat nga ada ti babuyan da ta narigat metlang nga mailako.*” They give their waste take outs to piggery owners to be used as a food for pigs. Leftover waste is hard to sell so they just give it.

This approach aligns with the Food Waste Hierarchy which emphasizes managing food waste through most sustainable practices. Composting and animal feeding are considered preferable to disposal, as they contribute to resource recovery and support a circular food system. In the study of Flying Ketchup (2024). This strategy is being practiced by several restaurant owners who are turning to waste segregation and composting, using food scraps and other biodegradable materials to produce compost for growing herbs and vegetables that

are later used in their dishes. This not only minimizes waste but also supports a sustainable, circular food system.

The following are examples of food establishments who are implementing composting. First is coffeehouse, implementing composting programs in countries such as the United State of America, and the Philippines. In these locations, the company composts used coffee grounds, food scraps, and other biodegradable waste. This initiative is part of Starbucks' broader sustainability efforts to reduce landfill contributions and promote responsible waste management. Through composting, Starbucks helps turn organic waste into nutrient-rich soil, supporting environmental conservation and urban agriculture. Second is Earth Kitchen, a farm-to-table restaurant in Quezon City, which practices composting by collecting their food scraps and sending these back to their partner organic farms. These food wastes are then reused as compost to enrich the soil and support sustainable farming. This approach reflects the restaurant's commitment to environmental sustainability and a circular food system. By reducing food waste and supporting organic agriculture, Earth Kitchen promotes responsible resource use and eco-friendly dining. Last is the Hillside Café and Juice Bar in Baguio City. This promotes zero-waste dining by composting its food scraps. The compost produced is then used for gardening and plant production within the café's space or through local partnerships. This practice supports their commitment to sustainability and environmental responsibility. By turning organic waste into useful compost, the café helps reduce landfill contributions and encourages eco-friendly habits among its staff and customers (Local Food PH, 2021).

Moreover, restaurants also implement *waste segregation*. Out of 20 respondents, 8 said that they are using this strategy, which helps streamline waste

management and promotes environmental responsibility. As defined by the Department of Environment and Natural Resources (DENR) (2023), waste segregation is the systematic separation of different types of waste materials at the point of origin to enable proper handling, recycling, composting, or disposal. This involves sorting waste into categories such as biodegradable, non-biodegradable, recyclable, and hazardous, which is essential for effective waste management and environmental protection.

It helps streamline disposal and lower waste management costs. Many establishments have implemented a three-bin system, designating separate containers for compostable materials and general waste. Respondents stated *“Agited kami ti basurahan para idia’y mabulbulok ken saan a mabulbulok nga basura, ken dagiti empleyado ti agdalu’ ken ag maintain kadagidia’y basurahan.”* The recyclable items can be reprocessed into new products, while general waste cannot be composted or recycled. This setup facilitates easier and more effective segregation at the source, ensuring that organic waste is properly composted, recyclable materials are recovered, and non-recyclable trash is disposed of appropriately. Another respondent stated, *“We make sure that we label our trash bins properly.”* They have labeled bins that separate the different kinds of waste like recyclable, compostable, and non-compostable. According to the Food Agriculture Organization (2022), biodegradable waste includes food scraps, vegetable peels, eggshells, used paper towels, and other organic matter that can naturally decompose. These materials are directed into clearly labeled bins designated for compostable items. On the other hand, non-biodegradable waste such as plastic wrappers, aluminum cans, and Styrofoam containers is separated and often sent for recycling or appropriate disposal, depending on local waste

management policies.

The following are examples of companies that are implementing waste segregation. First is Conti's Bakeshop and Restaurant located SM, Baguio City. This restaurant practices waste segregation by separating food waste, recyclables, and residuals in both their stores and commissaries. Employees receive training in waste classification to ensure that each type of waste is properly sorted and disposed of. The company also collaborates with eco-waste partners to facilitate the efficient collection and processing of segregated waste. This initiative reflects Conti's commitment to environmental responsibility and sustainable operations (Conti's Bakeshop, 2022). Second is The Wholesome Table, an organic restaurant chain in Metro Manila, which practices waste segregation by separating biodegradable kitchen waste for composting and sorting non-biodegradable waste for recycling. The restaurant collaborates with local organizations to implement zero-waste practices throughout its operations. This approach supports their commitment to sustainability, responsible waste management, and environmental stewardship (The Wholesome Table, 2020).

By clearly labeling and positioning bins, restaurants guide staff and customers to dispose of waste correctly, reducing contamination and improving overall waste diversion rates. Some restaurants use waste segregation with clear labeling and bin placement to simplify sorting, boost recycling rates, and ensure staff compliance, making waste management more efficient and eco-friendlier. As mentioned in the research of Villanueva and Navarro (2024) in urban Philippine restaurants, highlights how clear labeling and strategic placement of segregation bins encourage compliance among both staff and customers, leading to more effective waste diversion. Their study emphasizes that waste segregation

contributes to environmental conservation by minimizing landfill waste and supporting composting and recycling programs.

Out of 20 respondents, 12 said that they also implement *portion control* as a key strategy in their food preparation processes. As defined by the National Restaurant Association (2023), portion control refers to the practice of measuring and serving food in predetermined amounts to ensure consistency in serving size. They rely on standardized recipes. These are recipes that are detailed, tested recipes that restaurants use to ensure consistent quality, taste, portion size, and cost control every time a dish is made. A standardized recipe includes the exact ingredients, specific measurement, preparation steps in cooking method, and serving instructions. This ensures that whether the dish is prepared by a head chef or a new kitchen staff member, the result remains the same in flavor, texture, and presentation. They provide detailed instructions on the exact sizes, measurements, and quantities of ingredients to be used for each dish. As Naumov (2023) stated, Standardized recipes are essential for ensuring that food and drink are prepared consistently at a high quality. This consistency leads to increased consumer satisfaction as customers receive the same quality of food every time they order. This systematic approach to portioning supports efficient inventory use and contributes to sustainable kitchen operations.

A respondent stated, “*We include portion sizes in every preparation and plating guide so that even new staff can follow them accurately.*” Portion sizes are included in every preparation and plating guide to ensure consistency and help even new staff follow them accurately. By adhering to these recipes, restaurant staff can ensure that every serving is consistent in size and quality, which helps maintain customer satisfaction and brand reputation.

Based on the analysis of the researchers, defying the portion size can affect the taste, and customers tend to leave it over. For example, serving a combo meal which is composed of rice or potatoes, meats, vegetables, appetizers, and desserts, if the dessert has more than its normal portion size, it will be more appetizing to consume, and the consumer will have no more appetite to eat the rest of the meal. Accurate portioning contributes to cost savings by preventing overuse of ingredients. In line with the Sustainable restaurant practices, portion control minimizes food waste and improves the use of resources. Research by Santos and Villanueva (2024) also indicates that accurate portion control helps prevent under-serving, which can negatively impact customer satisfaction, as well as over-serving, which contributes to food waste and increased operational costs. Their study of Metro Manila restaurants showed that establishments implementing standardized recipes experienced significant reductions in food wastage and improved cost management. By reducing variability in serving sizes, restaurants optimize ingredient usage and improve inventory forecasting.

The following are examples of restaurants that are implementing portion control. First is Max's Restaurant in the Philippines. This restaurant implements portion control to ensure consistent serving sizes for popular dishes such as fried chicken, rice meals, and desserts. By standardizing portions, the restaurant effectively manages food costs and minimizes kitchen waste. This practice also helps maintain customer satisfaction by delivering uniform meal quality and presentation across all branches (Max's Group Inc., 2021). Second is The Bistro Group, which operates restaurants such as Italianni's, TGI Fridays, and Denny's in the Philippines.

It also implements portion control by using standardized recipe cards and

pre-weighed ingredients. This system ensures consistency in food quality, helps maintain brand standards, and supports efficient cost management. By accurately portioning ingredients, the group also reduces food waste and promotes operational efficiency across its various branches.

Out of 20 respondents, 6 said that they are implementing *inventory management*, which involves closely monitoring all ingredients to ensure efficient stock control. As defined by Smith and Taylor (2024), inventory management is the systematic process of ordering, storing, tracking, and controlling a company's inventory, including raw materials, finished goods, and supplies. It ensures that the right quantity of items is available at the right time, helping businesses avoid overstocking or stockouts, reduce costs, and maintain efficient operations. Some of the restaurants use inventory management to track stock levels, reduce waste, prevent shortages, and control food costs. It ensures smooth operations, accurate ordering, and better financial planning. One respondent stated they apply FIFO in managing their inventories. A respondent shared "*Kanayon mi nga i monmonitor dagijy nalaklaka nga ma decay tapno han lang nga agbalin nga basura. Ag purchase kami lang ti amo mi nga mausar t mysa aldaw tapno han kami nga ag over-stock.*" They monitor raw ingredients every day and they prioritize the goods that are easy to spoil to avoid waste from spoilage. They buy ingredients that can be used for a day to minimize over-stocking. The researchers observed that by regularly tracking stock levels, restaurant owners and staff gain valuable insights into the usage patterns of various ingredients, identifying which items are used most frequently and which have slower turnover rates. This detailed monitoring enables more accurate and data-driven purchasing decisions, reducing the reliance on rough estimates or guesswork. This practice helps avoid common pitfalls such

as overstocking, which can lead to spoilage, and understocking, which can disrupt service and customer satisfaction.

As a result, they can avoid overstocking or understocking, minimize food spoilage and waste, and optimize cash flow. Effective inventory management also supports menu planning and helps maintain consistent food quality by ensuring that fresh ingredients are always available when needed. These inventory management practices align with the sustainable restaurant practices by minimizing food waste through daily check and limited purchasing. Recent studies of (Delacruz & Mendoza, 2024) highlight that closely monitoring ingredient usage patterns enables restaurant managers to make informed purchasing decisions based on actual consumption data rather than relying on estimations.

The following are examples of restaurants that are implementing inventory management. First is The Coffee Bean & Tea Leaf, located in Metro Manila, which manages its inventory through a digital back-end system that tracks daily consumption and automatically triggers reordering when stock levels are low. Baristas are trained to monitor ingredient levels closely and use pre-measured portions during preparation to ensure consistency, minimize overuse, and prevent spoilage. This system supports efficient inventory control while maintaining product quality and reducing waste (The Coffee Bean & Tea Leaf Philippines, 2020). Second, Sushi Nori in Metro Manila utilizes digital inventory systems to monitor the usage of raw seafood and other perishable ingredients. By aligning their procurement with real-time sales data, they are able to reduce spoilage and accurately manage daily stock requirements. This approach enhances operational efficiency and supports the restaurant's commitment to maintaining freshness and

minimizing waste (Sushi Nori, 2021). Last, SaladStop in Manila, Philippines practices strict inventory management to handle the highly perishable nature of its fresh ingredients. The restaurant uses just-in-time ordering, real-time inventory tracking, and daily logs to monitor stock levels accurately. These strategies help ensure ingredient freshness, minimize food waste, and maintain consistent product quality across all branches (SaladStop! Philippines, 2021).

Some interviewees practice the strategy of *reusing or repurposing food scraps* to minimize waste and maximize resource efficiency in their kitchens. As defined by Williams (2025), reusing or repurposing food scraps involves the creative transformation of leftover, excess, or traditionally discarded food materials into new culinary products, such as stocks, sauces, garnishes, or even fermented items. This method not only minimizes food waste, but also encourages innovation in kitchens and supports sustainable practices in the food service industry. Some restaurants use reusing or repurposing food scraps to minimize waste, cut costs, and maximize resource efficiency. It supports by making full use of ingredients, reducing disposal needs, and creating value from items that would otherwise be discarded. Some respondents state “*Dagijy sobsobra nga ingredients ket ususaren dagijy staffs ko nga mainayun ijay kanen da.*” The excess ingredients are being used by the staff for their own foods. Commonly discarded parts such as vegetable peels, trimmings, and meat bones are collected and transformed into flavorful stocks and broths. These stocks serve as a rich base for soups, sauces, and stews, adding depth and complexity to the dishes while reducing the need for purchasing additional ingredients. Additionally, leftover rice is often repurposed by reheating or incorporating it into new dishes, which can be served either to customers or employees. This approach not only helps to decrease the volume of

kitchen waste but also enhances sustainability by utilizing every part of the ingredient, thereby contributing to cost savings and environmentally responsible kitchen practices. This strategy aligned with the Food Waste Hierarchy, which places reuse above recycling and disposal in waste management practices. By creatively repurposing food scraps, restaurants can use the surplus food effectively, extending the life of ingredients and minimizing unnecessary waste

Chefs are increasingly embracing a "root-to-stem" and "nose-to-tail" philosophy, ensuring that every part of an ingredient is utilized. For instance, carrot tops are repurposed into pesto, and meat trimmings find their way into broths and sauces, exemplifying a commitment to sustainability and innovation (Culinary Heritage, 2025). Leftover rice, a common byproduct in many restaurants, presents another opportunity for waste reduction. Rather than discarding surplus rice, establishments are finding creative ways to repurpose it into new dishes. For example, leftover rice can be transformed into fried rice, rice pudding, or even used as a filling for burritos, thereby minimizing waste and offering diverse menu options (Cookist, 2025).

The following are examples of companies that are implementing the reuse or repurposing of food scraps. First is Toyo Eatery in Makati City, which practices a sustainable kitchen philosophy by following a nose-to-tail and root-to-stem approach. The restaurant creatively repurposes food scraps such as vegetable peels, fish trimmings, and bones to make sauces, broths, and condiments. This method significantly reduces kitchen waste while showcasing the use of local and seasonal ingredients. (Toyo Eatery, 2021).

Through this practice, Toyo Eatery promotes both environmental responsibility and culinary innovation. Second is Hapag, located in Katipunan,

Quezon City, which embraces sustainability by creatively reusing meat and vegetable trimmings in their culinary process. These food scraps are transformed into rich sauces, pickles, and fermented components that are featured in their modern Filipino tasting menu. This practice not only minimizes kitchen waste but also highlights the restaurant's commitment to resourcefulness and innovation in Filipino cuisine.

Another strategy employed by the interviewee is the management of *takeout services* with a strong emphasis on reducing associated waste. As defined by Cruz (2024), takeout services refer to the practice in the food service industry where meals are prepared, packaged, and made ready for customers to consume off the premises. This system allows restaurants to extend their reach beyond dine-in customers, offering convenience, reducing wait times, and adapting to modern consumer demands such as online ordering and delivery. Restaurant owners gave importance to food packaging. Many restaurants have prioritized the use of eco-friendly, compostable materials for their takeout containers. They use eco-friendly, compostable packaging, and reusable take-out boxes as well. Some restaurants use takeout services with a focus on reducing waste to meet customer demand while promoting sustainability. A respondent stated, "*Na priorityk ti panang-urnos ti basura manipud takeout, isu nga nag-usarak iti eco-friendly a packaging ken mangted ti pagpilian kadagiti kostumer no kayatda wenno saan ti utensils, napkins, ken condiments.*" prioritized managing takeout waste by using eco-friendly packaging and offering customers the option to choose whether or not they want utensils, napkins, and condiments.

By using eco-friendly packaging and minimizing single-use items, they can attract environmentally conscious customers, cut costs, and reduce their

environmental impact. This shift not only helps to minimize plastic and non-biodegradable waste but also aligns with growing consumer demand for sustainable practices. Additionally, these establishments encourage customers to make conscious choices by offering the option to decline single-use utensils, napkins, and condiments when they are unnecessary. By empowering customers to reduce excess waste, restaurants foster environmental awareness and promote responsible consumption habits. Reyes and Villanueva (2023) states that this approach not only reduces unnecessary waste but also raises awareness among consumers, encouraging more responsible consumption habits. Their research highlights that such initiatives foster a collaborative effort between businesses and customers to minimize the environmental impact of foodservice operations.

The following are examples of companies who are implementing reusing or repurposing takeout services. First is Banapple Pies and Cheesecakes it is considered as a restaurant and it is located in Metro Manila and is well-known for its takeout-friendly meals and desserts, offering convenient options such as call-in orders, online ordering, and curbside pickups. These services allow customers to enjoy their favorite dishes with ease and flexibility, making Banapple a popular choice for on-the-go dining. The restaurant's commitment to accessibility and customer convenience has helped it maintain a strong presence in the takeout and delivery market. Similarly, Gerry's Grill, a well-known Filipino restaurant, offers take-out meals and party trays designed to cater to families and group gatherings. This service allows customers to enjoy traditional Filipino dishes conveniently at home or during special occasions, supporting both accessibility and customer satisfaction (Banapple, 2021).

In connection with food waste reduction strategies, there are techniques

used by the restaurant owners, such as proper chilling and repurposing. The researchers gathered data on how the restaurant owners process left-over foods at the end of the day. One of the effective techniques used by restaurant owners in managing leftovers is *chilling*, which plays a vital role in ensuring food safety and minimizing waste. As defined by the World Health Organization (WHO, 2023), chilling is the process of lowering the temperature of food to between 0°C and 5°C to slow down microbial growth and preserve freshness. This method is commonly used in food handling and storage to ensure safety, extend shelf life, and prevent spoilage, particularly for perishable goods such as meat, dairy, and prepared meals. A respondent stated, *“Before leftover food is stored, it undergoes a quality assessment to determine its eligibility and suitability for future use.”* Leftover food is carefully assessed for quality before storage to ensure it is still safe and suitable for future use. This evaluation involves checking for signs of spoilage, contamination, or compromised texture and flavor. Only food that meets the restaurant’s established safety and quality standards is selected for storage. Once verified, the leftovers are promptly placed in a chiller to maintain freshness and inhibit bacterial growth. Proper chilling preserves the food’s integrity and extends its shelf life without compromising safety. Typically, these leftovers are repurposed as meals for staff the following day, which not only reduces food waste but also contributes to cost efficiency. The Food Standards Agency (2025) emphasizes that chilling food correctly helps to stop harmful bacteria from growing. Foods such as cooked dishes and ready-to-eat items should be kept chilled to keep them safe. It is crucial not to leave these types of food standing around at room temperature. The agency recommends that food should be cooled as quickly as possible and then stored in a refrigerator set between 0°C and 5°C.

Proper chilling is just one of the techniques restaurants use, as chilling slows the spoilage of food. The researchers presented their ideas that using the suitable containers to store the leftovers, it maintains freshness and prevents contamination, ensuring that all ingredients and prepared foods are safely preserved for future use. Storing goods in a container requires clean, sanitized, and clearly labeled containers to maintain their quality and safety. Restaurants' daily routine for leftovers that still have quality is to immediately store and cool them in a designated container and make them available for employees to consume for the following days.

According to López et al. (2021), the use of designated containers for food storage not only prevents cross-contamination but also assists in inventory control and reduces food waste. The study highlights that well-structured storage protocols improve food reuse practices, particularly when leftovers are allocated for staff meals or repurposed in future dishes.

Another technique is *repurposing* leftovers, which involves transforming excess or unused ingredients from previous service periods into new dishes that are both appealing and profitable. As defined by D'Souza (2023), repurposing refers to the intentional reuse of materials, including food scraps, in a way that gives them a new purpose or extends their utility. In the context of foodservice, it involves converting surplus ingredients or trimmings such as meat bones, wilted herbs, or overripe fruits into useful components like stocks, marinades, and desserts, thereby enhancing sustainability and reducing operational waste. For example, roasted vegetables from dinner service may be blended into a soup for the next day's lunch menu, or unused cooked meats may be incorporated into wraps, fried rice, or pasta dishes. A respondent stated, "We *implement a practice*

of repurposing certain leftover ingredients to create new, exciting dishes, always ensuring that food safety and quality are never compromised.” repurpose select leftover ingredients into new dishes to reduce waste, while strictly maintaining food safety and quality standards. This practice enables restaurants to fully utilize their inventory, reducing spoilage and disposal costs. This process involves creatively transforming unused ingredients from previous service periods into new, appealing dishes. For instance, leftover roasted vegetables may be turned into soups, or excess cooked meats may be incorporated into wraps, pasta dishes, or fried rice.

This approach not only minimizes spoilage but also contributes to more efficient inventory management and cost control. These techniques which are proper chilling and repurposing are aligned with the Food Waste Hierarchy which prioritizes prevention, as they prevent edible food from becoming waste and promote efficient use of resources. According to Arsat et al (2025), repurposing edible leftovers as a strategy to reduce food waste, highlighting the creation of value-added products and the potential for appealing, profitable dishes. By repurposing these leftovers, we can bridge the gap between excess food and hunger, benefiting both the environment and the economy.

To understand the implementation of food waste management, the researchers gathered data on how restaurant owners implement their strategies. One of the fundamental food waste prevention strategies used in restaurant kitchens is the *First-In, First-Out (FIFO)* method. When restocking shelves, refrigerators, or dry storage areas, kitchen staff are trained to move older stock to the front and place newly received items behind it. This simple but effective practice promotes the consistent use of ingredients within their optimal shelf life,

especially for perishable goods such as dairy, produce, and meats. This practice is particularly critical for perishable items such as dairy, meats, and fresh produce, where shelf life is limited and the risk of foodborne illness from expired good is higher.

The researchers' gathered data on what steps the restaurant takes from the moment the ingredients are delivered to ensure minimal waste. Upon arrival, all ingredients are immediately labeled with the delivery date to track freshness. These are then stored according to the FIFO (First in- First out) method. Older ingredients are prioritized in cooking and food preparation, while new supplies are accurately organized and placed in proper storage to maintain order and accessibility. This method of inventory rotation helps reduce waste and maintain quality. According to Mettler (2022) emphasized that First in First Out aligns with the natural flow of perishable inventory in restaurants, reducing waste and spoilage. By using the oldest inventory first, restaurants can maintain product quality and ensure efficient use of resources. Regular physical inventory checks and accurate demand forecasting further enhance the effectiveness of FIFO in inventory management.

Restaurant owners also implement the strategy of composting as another effective method used to manage food waste. *Staff training and compliance* play a crucial role in the successful implementation of composting programs in restaurants. As defined by Rivera (2024), staff training and compliance refers to the ongoing process of educating employees on workplace policies, operational procedures, and regulatory standards, and ensuring that these guidelines are consistently followed. In the foodservice industry, it includes training staff in hygiene, food safety, waste segregation, and sustainability practices to maintain

operational efficiency and meet legal and quality standards. A respondent stated, *“we train our staff on what can and can’t be composted and during closing shifts, we do check the bins to ensure the right items.”* By educating employees on which materials are compostable, recyclable, or destined for landfill, restaurants can significantly reduce contamination in compost bins, which is essential for producing high-quality compost. Proper sorting minimizes the risk of rejected compost loads, which can be costly and counterproductive to sustainability efforts. Well-trained staff are also more likely to follow best practices consistently, ensuring that composting becomes an integrated part of daily operations rather than an afterthought. This level of organization not only supports the restaurant’s environmental goals but also enhances operational efficiency by reducing waste management issues.

Additionally, maintaining clear hygiene and cleanliness standards while composting helps keep the work environment safe and pleasant for both staff and customers. This approach supports the Sustainable Restaurant Practices which encourages the integration of environmentally conscious strategies within daily operations through systematic training and engagement of staff, and restaurants also enhance daily operational effectiveness. Nguyen and Patel (2020) highlight that well-trained staff are more likely to comply with composting protocols, which supports compliance with local waste regulations and elevates the restaurant’s public image as a sustainable business, it collectively emphasizes that continuous education and compliance monitoring are key drivers for successful food waste diversion through composting in the food service sector.

In addition to disposing of waste sustainably, portion control plays an important role in the strategy restaurants use to minimize food waste by carefully

limiting ingredients and using precise measurements. By using precise measurements and limiting ingredient use, restaurants can avoid over-preparing meals that might otherwise be thrown away. This reduces the volume of uneaten food, lowering the amount of waste sent to landfills or compost. It also helps maintain accurate inventory levels, reducing spoilage from unused ingredients. Consistent portion sizes improve customer satisfaction by ensuring predictable, high-quality dining experiences. Overall, portion control not only reduces waste but also supports cost savings, operational efficiency, and environmental responsibility. Studies show that by limiting ingredient use and applying precise measurements, restaurants can significantly decrease the amount of excess food prepared and discarded (Morris & Taylor, 2021).

Repurposing food waste resourceful practices are increasingly adopted by Restaurant owners to reduce overall waste and maximize ingredient use. For example, bones and trimmed vegetables that would normally be discarded are often used to make flavorful stocks or broths, adding depth to dishes while minimizing waste. Leftover rice is frequently recooked and transformed into new meals, providing cost-effective options for both employees and customers. Additionally, citrus peels are creatively repurposed into house-made syrups and garnishes, enhancing menu offerings with unique flavors. These practices not only reduce the environmental impact of food waste but also contribute to cost savings for the restaurant. By finding innovative ways to reuse food by-products, restaurant owners promote sustainability and demonstrate culinary creativity. Restaurant owners are increasingly adopting innovative strategies to repurpose food waste, thereby minimizing environmental impact and maximizing resource efficiency. According to Lopez and Henderson (2023), common practices include

transforming bones and vegetable trimmings into stocks and broths, which add flavor while reducing waste. Similarly, Martinez et al. (2022) highlight that leftover rice and other surplus ingredients are often reused in new dishes or staff meals, enhancing cost efficiency without compromising quality

Implementation of *proper storage* of goods is also essential for minimizing food waste in restaurants by preventing spoilage. As defined by Gonzales (2024), proper storage refers to the correct handling, placement, and preservation of food and materials under appropriate conditions such as temperature, humidity, and packaging to maintain quality, prevent spoilage, and ensure safety. In restaurant operations, proper storage includes labeling, using sealed containers, and separating raw from cooked items to comply with food safety standards. An interviewee said, “ *By properly storing all the ingredients with a clear label,*” Labeling items clearly with their expiration dates helps staff quickly identify what needs to be used soon, preventing unnecessary waste. Proper storage also maintains the quality and safety of food, which is critical for customer satisfaction and health standards. Overall, these practices contribute to efficient inventory management and support sustainability by reducing food loss. Proper storage of food supplies is a critical strategy for minimizing food waste in restaurants across the Philippines. Studies have shown that Filipino restaurants adopt organized storage techniques to reduce spoilage, which is a significant contributor to food loss in the country (Delos Reyes & Santos, 2022).

In addition to that, *Monitoring or checking* the stock level daily to identify the usable ingredients and wastes. As defined by Delos Reyes (2024), monitoring or checking is the systematic process of observing, recording, and evaluating tasks or conditions to ensure they meet required operational, safety, or quality standards.

In restaurant operations, it involves checking storage temperatures, expiration dates, cleanliness, and staff compliance to prevent errors and uphold service quality. This proactive approach limits unnecessary purchasing and helps identify ingredients nearing expiration (Smith & Lee, 2020).

Implementing *proper waste segregation* is essential for effective waste management in restaurants. Bolt (2024) defined waste segregation as a method of identifying, classifying, dividing, and sorting garbage and waste products to encourage material reduction, reusing, and recycling. As part of an increasingly vital strategy in sustainable foodservice operations, *food waste segregation* has become a central focus for many restaurants. As defined by Martinez (2024), food waste segregation is the systematic separation of food waste into distinct categories such as biodegradable, non-biodegradable, and recyclable materials at the point of disposal. This practice enables proper handling, composting, recycling, or disposal, and is essential in promoting sustainable waste management in foodservice establishments. The restaurant owners stated “*By separating liquid waste such as sauces, soups, ketchups, and oils from solid waste, restaurants can prevent contamination and make recycling or composting processes more efficient.*” Liquid waste is typically poured down the sink or into designated containers, while solid waste is disposed of in specific garbage bins. This practice reduces the risk of clogging waste streams and ensures that compostable or recyclable materials remain uncontaminated. Proper segregation also helps restaurants comply with local waste disposal regulations and promotes environmental sustainability.

Overall, a clear separation of liquid and solid waste improves operational hygiene and supports more effective waste diversion efforts. A study by Balaria et

al. (2021) revealed that 80% of restaurants and eateries in Cabanatuan City disposed of used cooking oil directly into the kitchen sink, indicating a need for improved education and infrastructure to manage liquid waste properly. These findings underscore the necessity for continuous training, infrastructure development, and policy enforcement to enhance waste segregation practices in the Philippine restaurant industry.

Furthermore, the researchers gathered data on how the restaurant owners segregate their food waste. Staff play a vital role in ensuring an effective and sustainable waste management system within restaurants. By accurately sorting waste into biodegradable and non-biodegradable streams, they help prevent contamination that can compromise recycling or composting efforts. Proper categorization at the source increases the chances of waste being processed correctly, reducing the volume sent to landfills. This not only supports environmental goals but also helps restaurants comply with local waste regulations. Well-trained staff contribute to smoother operations by reducing the need for re-sorting or correcting mistakes later in the process. Ultimately, their consistent efforts improve overall efficiency, cleanliness, and sustainability in the restaurant's daily operations. Ghinea & Ghiuta (2021) emphasize that restaurants that implement clearly labeled compost bins and ensure source-level segregation reduce contamination in the waste stream, allowing for more effective organic waste processing. Meanwhile, non-biodegradable items such as plastics, aluminum cans, and Styrofoam containers are separated for recycling or proper disposal, depending on municipal guidelines.

In addition to employee training, restaurants enhance their waste segregation practices by providing clearly labeled waste bins, which serve as

visual cues to guide proper disposal behaviors. Each trash bin is marked with descriptive labels and, in many cases, color codes that indicate the type of waste it is intended to collect: recyclable materials, compostable food waste, or general refuse. These bins are strategically placed in high-traffic and operationally critical areas such as kitchen prep stations, dishwashing areas, and service zones. This ensures that staff can easily access the appropriate bins without interrupting workflow, thereby integrating waste segregation seamlessly into daily routines. The clear labeling of bins not only aids staff but also assists any additional personnel, such as cleaning crews or part-time workers, in correctly identifying and sorting waste at the point of disposal. This reduces cross-contamination between waste streams, a common issue that can render recyclable or compostable materials unusable if not handled correctly.

As mentioned by (Zhu & Li, 2021) Effective waste segregation in restaurants depends not only on employee training but also on the implementation of clearly labeled waste bins, which act as visual guides to promote correct disposal behaviors. Labeling, often accompanied by color-coding systems, simplifies the identification of appropriate disposal methods for recyclables, compostables, and general refuse, helping to reduce user error and contamination between waste streams.

Also, many restaurant staff emphasized the importance of using *color-coded and clearly labeled bins* to make waste segregation easier and more accurate. As defined by Villanueva (2024), color-coded and clearly labeled bins refer to waste containers that are assigned specific colors and marked with visible labels to identify the type of waste they are meant to collect, such as biodegradable, non-biodegradable, recyclable, or hazardous materials.

This system improves waste segregation efficiency, supports environmental compliance, and reduces contamination in disposal processes. These visual tools help reduce confusion, especially for new or busy employees. Color codes and labels act as quick guides, improving sorting and reducing mistakes during peak hours. Interviewee stated, “*Color-coded bins with simple instructions or visual guides can help reduce mistakes in waste segregation, especially for new or busy staff.*” using color-coded bins with clear instructions or visual guides helps minimize mistakes in waste segregation, particularly among new or busy staff members. They also serve as simple training aids and help promote a shared sense of responsibility among all staff. Color-coded bin systems typically assign green to compostable items, blue to recyclables, and black or gray to general waste. These visual markers help staff quickly distinguish between waste categories, even during fast-paced service periods. Research by Wang et al. (2023) emphasizes that such visual systems support behavioral change by making correct actions easier to follow, even with minimal formal training.

Segregating waste is essential to *prevent contamination* that can compromise the safety and quality of consumable goods. As defined by Ramos (2024), prevent contamination refers to the practice of implementing measures to stop harmful microorganisms, chemicals, or foreign objects from coming into contact with food, equipment, or surfaces. In foodservice settings, this includes proper handwashing, separation of raw and cooked items, sanitation of tools, and use of protective gear to ensure food safety and quality. When biodegradable and non-biodegradable waste are not properly separated, harmful bacteria or chemicals can transfer to food preparation areas or storage, increasing the risk of foodborne illnesses. Proper segregation also ensures that recyclable and compostable

materials remain uncontaminated and usable, supporting both hygiene and sustainability in foodservice operations. Contamination occurs when organic waste is stored alongside packaging materials, plastics, or cleaning chemicals, potentially exposing consumable products to pathogens or toxic residues (Kibert et al., 2020). This is particularly concerning in restaurants, where the cross contact of waste with surfaces used for food preparation can result in foodborne illnesses, spoilage, or violations of public health regulations (Martínez-Sánchez et al., 2021).

After knowing the strategies and techniques implemented by the selected restaurant owners, the researchers also presented inevitable challenges that were encountered during the implementation. These are the following data.

Out of 20 restaurants 17 of them faced the issue of mixing *old and new stock*. As defined by Villamor (2024), mixing of old and new stock is the improper inventory practice of combining older products with newer ones without following stock rotation methods such as FIFO (First In, First Out). This results in older items being left unused, increasing the risk of spoilage, food waste, and non-compliance with food safety standards. Respondent 1 said “*During busy hours, it's easy for staff to grab whatever is most accessible, which sometimes leads to newer stock being used first and older items going to waste.*” Respondent 2 also stated “*Nadadaganan ang mga dating naka lagay sa freezer dahil dun napagmimix ang old at new stock.*” same with respondent 3 who said, “*The older stock is tampered with the new ones that why sometimes instead of the older stock we use the new ones.*” Based on these responses, this gives the idea to researchers that proper stock rotation is essential in preventing food waste and maintaining product freshness in restaurants. When staff are trained to follow the “first in, first out” (FIFO) method, older inventory is used before newer items, reducing the risk of

spoilage. An example is the chains like Pizza Hut as part of a huge franchise system; Pizza Huts rely on an enormous supply chain to standardize their food. Pizza Hut ensures constant stock rotation by freezing and packaging ingredients, then applying FIFO practices in storage and preparation to maintain freshness across locations.

Avoiding the mixing of old and new stock ensures that food nearing its expiration date is prioritized, helping to minimize unnecessary waste. This practice also supports better inventory control, as it becomes easier to track usage patterns and reorder supplies more accurately. During busy hours, maintaining organized storage and clear labeling can help staff quickly access the correct items without compromising efficiency. Overall, consistent stock rotation improves food safety, reduces costs, and contributes to more sustainable restaurant operations. According to Stevenson (2021) mixing new and old stock without a systematic approach can lead to inventory imbalances, and product deterioration. Failure to rotate stock properly can compromise safety and regulatory compliance. Limited storage space and unclear labeling further contribute to the confusion, causing products to be pushed to the back and forgotten. New or untrained employees may also make mistakes in stock rotation.

Out of 20 restaurants, 8 of have concerns about *cross-contamination*. As the Food Standards Agency, (2024) notes cross cross-contamination is what happens when bacteria or other microorganisms are unintentionally transferred from one object to another. The most common example is the transfer of bacteria between raw and cooked food. Compost bins, if not emptied regularly, can produce unpleasant odors and attract pests. A respondent stated, “*Some staff may forget or rush when disposing garbage, leading to cross-contamination of waste.*”

It takes ongoing reminders, monitoring, and refresher training to keep the system working properly". When employees fail to properly segregate waste, compostable materials can become contaminated with plastics or other non-compostable making the entire batch unusable. Ignoring labels or mixing waste out of habit can lead to serious setbacks in sustainability efforts. Proper training and regular reminders help staff stay mindful of correct procedures, even under pressure. Clearly labeled bins and easy-to-follow systems can reduce errors and encourage consistent compliance. Preventing cross-contamination not only supports environmental goals but also maintains hygiene and operational efficiency in the restaurant.

It was mentioned by the restaurant owners that *incorrect measurement of ingredients is also an issue in food preparation*. As defined by Navarro (2024), incorrect measurement of ingredients refers to the inaccurate weighing or portioning of food components during preparation, which can lead to inconsistent food quality, increased food waste, and poor inventory control. In restaurant operations, this issue often results from a lack of standardized procedures or improper staff training. Incorrect measurements occur due to last-minute changes or specific customer expectations. These unexpected adjustments can lead to over-preparation or the creation of oversized portions that exceed what is needed. A respondent stated "*No masubraan dagijy ingredients, ada nga ada ti maibasura.*" When this happens, extra ingredients may be used unnecessarily, increasing the chance of food going to waste. Unused prep items, especially perishable ones, may not always be stored properly or reused in time, leading to spoilage. Careful management and communication among staff are essential to minimize these issues and ensure ingredients are used efficiently. By planning and maintaining

flexibility without compromising portion control, restaurants can better manage waste and reduce unnecessary loss. According to Brown, & Davis (2021) accurate measurement in food preparation is essential in restaurants to ensure consistent product quality, cost control, and customer satisfaction. Incorrect measurement of ingredients can lead to variations in taste, portion sizes, and nutritional content, ultimately affecting the restaurant's reputation and profitability.

The *lack of time* during busy hours makes it difficult to collect reusable scraps or leftovers, and some are simply too damaged to repurpose. As defined by Cruz (2024), lack of time refers to the insufficient availability of time to properly perform tasks, follow procedures, or complete responsibilities, often resulting in rushed work, skipped steps, or errors. In foodservice settings, this can lead to poor food handling, improper waste segregation, or failure to follow safety protocols during peak hours. The fast-paced environment prioritizes speed and service, making it difficult to focus on separating items that could be repurposed. As a result, many potentially reusable ingredients are discarded instead of being saved.

Additionally, some leftovers are too damaged or contaminated to be reused safely or effectively. This leads to increased food waste and misses opportunities for more sustainable practices. According to our interviewee, “*minsan nakakalimutan ng mga staffs ko yung mga kanin na pwede pa nilang gamitin na nasa ref kaya nababasura na lang.*” Occasionally, staff forget about leftover rice stored in the fridge to repurpose, and that could go to waste. Bilska et al (2022) state that the fast-paced environment in kitchens and dining areas can prevent staff from collecting and repurposing food scraps. The need to maintain service speed and efficiency often takes precedence over waste reduction efforts. As a result, potentially reusable food items may be discarded rather than utilized for other

purposes.

Improper waste segregation remains a challenge for many restaurants, especially when kitchen space is limited. As defined by Delgado (2024), improper waste segregation refers to the incorrect or careless separation of waste materials, where biodegradable, non-biodegradable, and recyclable items are mixed. This practice hampers effective waste disposal, increases environmental risks, and prevents proper recycling or composting efforts, especially in foodservice establishments. The interviewee mentioned “*Narigat nga maisina dagijy basura nu bassit lang t basuraan.*” The lack of room for multiple separate bins makes it difficult for staff to sort waste efficiently.

This often leads to materials being disposed of incorrectly, increasing contamination and reducing the effectiveness of recycling or composting efforts. Additionally, they also said “*Dagijy metlang gamin customer ket sabali pangibasuraan da uray naka indicate en ti type ti basura, isu rason nga marigatan dagijy staff nga mangiseparate.*” Customers frequently mix different types of waste in the wrong bins, creating more work for staff and further complicating the segregation process. These challenges highlight the need for better waste station design, clear signage, and customer education. Improving both staff workflow and customer cooperation can help overcome space constraints and support more effective waste management practices. According to Rios et al (2022) foodservice businesses are often unwilling to invest in waste management systems even though they could cut operating costs by reducing perishable loss existing research often includes other sectors of the food and beverage industry, such as food producers, manufacturers, and this has left the foodservice sector with a comparative lack of initiatives and knowledge on waste management, and food

managers are consequently required to renew their managerial systems.

In busy restaurant settings, staff often struggle to maintain *proper waste segregation* during peak hours. As defined by Lopez (2024), proper waste segregation is the systematic process of separating waste into categories such as biodegradable, non-biodegradable, recyclable, and hazardous materials at the point of disposal. This practice promotes efficient waste management, reduces environmental pollution, and enables effective recycling and composting in both households and food establishments. When overwhelmed by the volume of customers, employees may resort to mixing all types of waste, biodegradable, recyclable, and non-recyclable into a single bin. This practice compromises the restaurant's sustainability goals and disrupts the effectiveness of composting and recycling systems. Mixed waste contaminates otherwise usable materials, making them unsuitable for recovery or reuse.

It also increases the amount of waste sent to landfills, which contributes to environmental degradation. To prevent this, restaurants need to reinforce waste management protocols and provide adequate staffing or support tools during high-traffic periods. Studies of Martinez & Nguyen, (2023) indicate that restaurant staff often combine different types of waste such as organic, recyclable, and general trash due to time constraints, limited resources, or lack of training.

Additionally, a restaurant owner also observed that *unfinished meals* and customers declining to take leftovers home contribute to food waste. As defined by Reyes (2024), unfinished meals refer to portions of food that have been served to customers but are not fully consumed, often left on plates and discarded after dining. A respondent stated, “*There are customers who cannot finish their food because of a lot of serving.*” Some customers are unable to finish their food due to

overly large servings, which can lead to increased plate waste and the need to reassess portion sizes. This type of waste commonly results from oversized portions, customer preferences, or lack of appetite, and significantly contributes to post-consumer food waste in restaurants. A restaurant owner observed that unfinished meals left by customers are a notable source of food waste. Often, large portion sizes or changing appetites lead to diners leaving significant amounts of food on their plates. For example, combo meals often come with large servings. As a result, some customers are unable to finish their food. Many customers also decline to take leftovers home, even when offered takeout containers. This results in perfectly edible food being thrown away at the end of a meal. Encouraging smaller portions and promoting leftover takeaway options can help reduce this type of waste.

According to Fajriah et al. (2024) Customers leave food on their plates for various reasons, making it difficult to repurpose the leftovers, several factors influence the amount of food waste at the consumer consumption, mostly related to consumer behavior, such as over-ordering excessive purchasing, internal cultural values, dining purposes, and food consumption practices.

Another issue is *inventory control*, such as unexpected delivery delays and irregular stock checks. As defined by Mendoza (2024), inventory control refers to the systematic process of monitoring, managing, and regulating stock levels to ensure the right quantity of items is available when needed while minimizing overstocking, spoilage, or shortages. In foodservice operations, effective inventory control helps maintain product quality, reduce food waste, and optimize overall cost efficiency. An interviewee mentioned “Without *regular checks*, *expired items might stay unnoticed*.” Regular checking, expired items may go unnoticed,

leading to food safety risks, quality issues, and increased waste, which is why consistent inventory monitoring is essential for maintaining freshness, protecting customers, and minimizing costs. This can also lead to forgotten or expired items, especially when staff are overwhelmed or rushing during peak times. A respondent stated, “Unexpected *delivery delays throw off our inventory plans.*” Unexpected delivery delays disrupt our inventory plans by causing shortages, affecting meal preparation, and increasing the risk of food waste from overcompensating or rushing orders. When deliveries arrive late or stock isn't checked regularly, it becomes easy for ingredients to be misplaced or forgotten. Additionally, they also said, “Staff *can still make mistakes in rotating stock, especially if they're new or rushing during busy times.*” Staff can sometimes make mistakes in rotating stock, particularly when they are new or working under pressure during busy hours, which can lead to spoilage and increased food waste. During busy periods, overwhelmed staff may overlook older items, leading to expiration and unnecessary waste. Poor inventory tracking also makes it harder to plan menus and manage supply levels effectively. Consistent stock monitoring and organization are essential to prevent spoilage and reduce food waste. According to Posease (2024) many restaurants do not consider seasonal changes, holidays, and special events when they plan their inventory. This can lead to excess stock or a shortage and both can affect profits.

The researchers found that customer behavior and preference significantly contribute to food waste in restaurants. Restaurant owners have noticed that a considerable amount of food waste comes from *customers not finishing their meals*. As defined by Alvarez (2024), customers not finishing their meals refers to the situation where diners leave portions of their served food uneaten, often

resulting in plate waste. This issue is typically caused by oversized servings, dissatisfaction with food quality, changing appetite, or distraction, and contributes significantly to post-consumer food waste in restaurants.

One respondent stated “*adda ngay jay dadduma nga customer nu agorder ti combo meals ket saan da nga maibus ta adu gamin.*” and another respondent also said “*Maraming order yung customer pero di niya kayang ubusin.*” Some customers find it difficult to finish large portions like combo meals, which often leads to food waste. A major reason for this is over-ordering, where customers request more dishes than they can consume. In other cases, the food may not meet customer expectations, whether due to taste, presentation, or overall satisfaction, leading them to leave it uneaten. Some customers may also have health or dietary restrictions, or simply smaller appetites, which limit how much they can eat. To address this, restaurants can consider offering flexible portion sizes, clearer menu descriptions, and staff training to guide customer choices more effectively. These statements of the respondents suggest their positive attitude in ordering more food, but they lacked the intention to consume everything they ordered. This over-ordering behavior relates to the perceived behavioral control on the Theory of Planned Behavior. (Freedman & Brochado, 2019) present the ideas that have consistently shown that large servings can result in customers being unable to finish their meals, especially in full-service restaurants.

Garnishes are often left untouched on plates, resulting in a common source of food waste in restaurants. John (2023) refers to garnish as the decorative elements that are added on top of a dish or drink. Many customers view garnishes as purely decorative rather than edible components of the dish. Examples of this are garnishes that may go unconsumed could include as herbs, lemon wedges, or

decorative vegetable slices. These items, while intended to enhance a dish, can frequently be discarded, contributing to overall food waste in households. Some diners are simply not accustomed to eating garnishes, especially if they are unfamiliar or culturally uncommon.

A frequently cited reason is that the taste of the garnish is not appealing or does not complement the main dish. A respondent stated, “*Customers reject certain sides or garnishes, leading to waste.*” Customers sometimes reject certain sides or garnishes, which often go untouched and end up as food waste despite being perfectly edible. As a result, even fresh and edible garnishes are discarded, contributing to unnecessary waste. This may be because the customers don't like to eat it, or some customers don't eat it as they think it's just for aesthetics. Another reason is that the people here in Cordillera are not accustomed to eating garnishes. These reflect the subjective norms and cultural attitudes in that place about how they view what is edible and necessary to consume as connected to the Theory of Planned Behavior. To address this, chefs could consider using edible garnishes that enhance flavor or incorporating them into the dish in a more appealing way to encourage consumption. Studies of (Filimonau et al., 2020) have also shown that cultural food habits influence garnish consumption; in some regions, diners are not accustomed to eating garnishes, considering them purely aesthetic.

Many restaurant customers often leave behind leftovers, *choosing not to take them home* despite being offered the option. As defined by Ramirez (2024), choosing not to take them home refers to the decision of customers to leave leftover food at the restaurant instead of requesting it to be packed for takeout. This behavior contributes to post-consumer food waste and may be influenced by

cultural norms, embarrassment, convenience, or lack of awareness about food waste reduction. A respondent stated, “*Some of the customers don't want to take home the leftovers.*” Some customers choose not to take home their leftovers, which unfortunately contributes to avoidable food waste despite efforts to offer sustainable options. This behavior suggests a social or personal reluctance to carry unfinished food, possibly due to concerns about food safety, convenience, or stigma. Some customers feel that taking leftovers is unnecessary or inconvenient, especially if they are not heading directly home. Others believe the remaining food is too small in portion to be worth saving. As a result, perfectly edible food is frequently discarded, contributing to overall food waste in the restaurant. According to Kim and Lee (2022) despite awareness of food waste issues, a significant portion of diners declined takeout options, citing concerns over food safety, social stigma, and personal habits. Their study revealed that the preference to leave leftovers behind is often driven by a combination of cultural attitudes and perceived inconvenience associated with carrying food.

The researchers gathered data on how the restaurant provides *training* that would help them improve food waste prevention and proper handling. As defined by Bautista (2024), training refers to the organized process of teaching employee’s specific knowledge and skills required to perform their job effectively and comply with workplace standards. In the foodservice industry, training includes topics such as food safety, hygiene, waste management, portion control, and customer service to improve performance and ensure consistent quality. A respondent said, “*Staff are regularly trained by their supervisor on food waste reduction, hygiene, and proper waste disposal, including how to separate organic from non-organic waste and encourage guests to take home leftovers.*” These trainings occur

monthly and during onboarding, supported by hands-on demonstrations to ensure a clear understanding.

According to Martins and Pereira (2022), regular and structured training programs for foodservice staff significantly improve waste handling efficiency and compliance with sustainability goals. Monthly training sessions and onboarding programs that focus on waste segregation, food storage, and spoilage prevention are particularly effective in ensuring long-term staff engagement and behavioral change.

New employees receive *orientation* on proper food handling to ensure safety and reduce waste. As defined by Domingo (2024), orientation refers to the initial process of introducing new employees to the workplace environment, policies, procedures, and expectations of an organization. In foodservice settings, orientation often includes familiarizing staff with kitchen operations, safety protocols, waste segregation, and customer service standards to ensure a smooth transition into their roles. A respondent said that *“Dagijy leaders mi isuda ti mapmapan maki seminar or maki orientation idjy events santo da e-update no anya man ti na seminar da and e-orient da met dakami ken dagijy staffs mi”* their leaders are the ones who are attending seminars and orientations, and they share it to their staff. They are also trained in inventory management to help track and use ingredients efficiently. One important practice taught is the use of “Use First” shelves, which prioritize items that are close to expiration. Staff are given walkthroughs of composting areas to understand how to properly separate food waste.

They also tour storage areas to learn best practices for organization and maintaining freshness. As emphasized by Nguyen and Ramos (2021), these

practices not only help reduce waste but also promote accountability and food safety within kitchen operations.

During shifts, managers and supervisors give regular *reminders* to their staff on sorting food waste properly by continuously guiding them on waste segregation and by regularly reminding them that if the trash is full, they must change it immediately. As defined by Fernandez (2024), reminders refer to verbal, written, or visual prompts used to reinforce important tasks, procedures, or behaviors within the workplace. A respondent said “*By regularly reminding them if the trash has been full, they change it immediately, by demonstrating them first in properly handling correctly in their first day of work in us* ” ensures proper waste management by demonstrating correct trash handling on their first day and regularly reminding staff to change bins immediately when full. In restaurant operations, reminders may include signage about handwashing, food safety guidelines, or waste segregation instructions to help staff maintain consistency and compliance with operational standards. The restaurant's owner or supervisor *demonstrates* them first in properly handling correctly and handling near-expiry items, with monthly refreshers to maintain consistency. As Chaturvedi (2025) stated, encouraging a mindfulness about waste management starts with leadership. Demonstrations, visual reminders, and incentives for waste reduction can help staff stay committed to sustainability efforts. Simple actions like correctly measuring ingredients, using food scraps creatively, and handling leftovers properly can make a big difference.

Clear instructions are communicated to staff both verbally and through posted signs throughout the restaurant. As defined by Navarro (2024), clear instructions refer to the delivery of detailed and unambiguous guidance that

enables staff to understand and carry out their duties accurately and efficiently. In restaurant environments, this helps reduce errors, improve food safety practices, and ensure that employees follow proper procedures, especially during busy or high-pressure shifts. A respondent stated, “*We instruct the staff about the proper waste prevention lalo na sa segregation*”. We instruct our staff on proper waste prevention, especially in segregation, to ensure that recyclables, compostables, and trash are correctly sorted and managed. These guidelines serve as constant reminders to follow proper procedures for reducing food waste. Visual cues help reinforce training and make it easier for staff to act quickly and correctly, even during busy times. This consistency helps ensure that all team members contribute to waste reduction efforts. Clear communication supports an environmentally responsible and efficient operation.

According to Luu (2020) communication plays a crucial role in influencing employees' intention and behavior to reduce food waste, suggesting that clear verbal instructions and posted signs can enhance consistency in reducing food waste and promote environmentally responsible operations among hospitality staff. Effective communication can lead to a significant reduction in food waste behavior among hospitality employees.

The respondents also identified the kinds of support and resources coming from outside or within the restaurant that would help restaurants improve their food waste management. According to a respondent, one form of support is attending training sessions outside, which they then apply and share the learnings with their staff. An example of this is GypsyBaguio by Chef Waya. The restaurant participates in WWF-Philippines' The Sustainable Diner program, which provides external training focused on food waste measurement, separation, and sustainable

kitchen practices. Following the training, GypsyBaguio implemented an internal workshop for kitchen staff to share what was learned. Staff were trained to identify sources of food waste, track leftovers and trimmings, and adjust food preparation processes accordingly. This strategy enabled the restaurant to reduce unnecessary waste, make informed procurement decisions, and embed sustainability into its daily operation (WWF-Philippines, 2021)

Another support is *choosing reliable and consistent suppliers*. The Restaurant owners also choose to partner with reliable and consistent suppliers of ingredients used in their restaurants. It is to ensure that they help deliver fresh and high-quality products. According to Jayatellika (2024) supplier reliability management is crucial for overall product reliability, focusing on educating suppliers about customer reliability requirements and ensuring parts meet these requirements through verification and validation. This process elevates supplier reliability beyond mere dependability, enhancing supply chain effectiveness.

Furthermore, *choosing reliable and consistent suppliers* is essential to a restaurant's strategy for reducing food waste and maintaining high-quality service. As defined by Lim (2024), choosing reliable and consistent suppliers refers to the selection of vendors who can regularly deliver high-quality products on time and in proper condition, ensuring smooth operations and food safety in restaurants. This practice helps prevent spoilage, supports inventory accuracy, and maintains the quality of dishes served to customers. Dependable suppliers ensure that ingredients arrive fresh and within the expected timeframe. This reduces the chances of spoilage caused by delayed or poor-quality deliveries. Timely deliveries also support better inventory planning and minimize overstocking. A respondent stated "*Reliable supply chains lead to smoother operations and less*

last-minute scrambling.” Reliable supply chains ensure smoother operations by minimizing last-minute issues and helping maintain consistent inventory and service quality. Strong supplier relationships contribute to sustainability and the overall success of the business. As emphasized by Marijn Overvest (2025) Setting well-defined criteria is crucial for effective supplier selection, as it can lead to improved procurement operations, cost efficiency, and minimize risk of receiving poor- quality ingredients, further supporting waste reduction and food safety practices.

One manager explained that having fresh ingredients delivered on time, this reduces the risk of spoilage upon delivery and supports their effort to minimize trustworthy suppliers allows them to plan their inventory more accurately, which reduces overstocking and unnecessary waste. Another emphasized that consistent deliveries also support smooth kitchen operations by preventing delays and shortages. Suppliers who follow agreed standards in packaging and handling also help maintain food safety and reduce damaged goods. In the long run, building strong supplier relationships contributes to both sustainability goals and overall business efficiency.

Benchmarking is another kind of support that provides valuable insights into operational efficiency by enabling restaurants to compare their performance against industry standards. As defined by Torres (2024), benchmarking refers to the process of comparing an organization’s performance, practices, or standards with those of leading or best-performing establishments in the same industry. In the foodservice sector, benchmarking helps identify areas for improvement, adopt best practices, and maintain competitive quality in operations, sustainability, and customer service.

A respondent stated, *“We use this because benchmarking data helps us to monitor and compare our food waste levels against industry standards, and this helps where or can reduce waste and improve our sustainability practices in food waste management.”* Use benchmarking data to monitor and compare our food waste levels with industry standards, helping us identify areas for improvement and enhance our sustainability practices. While some establishments utilize benchmarking to track food cost percentages, labor efficiency, and customer satisfaction ratings, others employ it to monitor food waste levels and assess sustainability practices. Garcia & Tan (2022) found that by identifying peak dining hours and popular menu items, restaurants can better align food preparation with demand, thus reducing overproduction and minimizing waste. However, not all restaurants utilize benchmarking data due to limitations in time and resources. In some cases, food waste is managed through observation rather than precise measurement, particularly during high-volume operations when tracking becomes impractical.

CHAPTER IV

CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the conclusions and recommendations of food waste management strategies of restaurants in Central Business Districts of La Trinidad, Benguet.

Conclusions

Based on the results of the discussions and interpretations conducted to this study, the researchers arrived at the following conclusions.

1. The restaurant owners or managers from the selected restaurants in La Trinidad, Benguet are implementing food waste management strategies such as reducing food waste and balancing inventory levels.
2. The restaurant owners or managers from selected restaurants in La Trinidad, Benguet, are experiencing challenges in managing their food waste.
3. Based on the findings, aside from the training given to the employees, restaurant owners or managers from these selected restaurants in La Trinidad, Benguet depend on the support and sources coming from reliable suppliers and through benchmarking.

Recommendations

Based on the conclusions of this study. Recommendations are drawn for restaurants in La Trinidad, Benguet that can help them address food waste management.

1. The restaurants in La Trinidad, Benguet should try to share food waste

management strategies with some food establishments.

2. It is recommended for the restaurant owners to conduct a review of the strategies and do an assessment for the strict implementation of these strategies.
3. It is recommended to have a tool or support within strategic management, including benchmarking and having dependable suppliers.

Other Recommendations

4. To further improve food waste reduction, restaurants can start by reviewing and adjusting their menus, particularly items that regularly result in excess waste, which should be modified or replaced.
5. It is recommended that restaurant owners conduct regular food waste audits that could help identify waste patterns and areas for improvement.
6. Collaboration with local governments can offer access to training, subsidies, and other forms of support.
7. It is recommended for future researchers that this study will be continued for more comprehensive study.

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APPENDICES

CONSENT FORMS

	King's College of the Philippines Research Ethics Committee
	Approval Letter

Date: 25 February 2025

KCP-REC Application Serial Number: **023s2025**

*Name of the Principal Researcher: **Benjie Ford B. Mocati***

*Address of the Principal Research: **Shilan, La Trinidad, Benguet***

Title of the Proposal: “Food Waste Management Strategies of Restaurants in Central Business Districts of La Trinidad, Benguet”

Researchers: *Benjie Ford B. Mocati*
Rose ann P. Binwihan
Marielle T. Kilip
Marjon S. Quilladas
Rhea lee T. Tay-ew
Jeyzelie J. Terrado

Dear Mr./Ms. *Benjie Ford B. Mocati*

Re: Approval Letter

The following documents were duly received by this office from you on the said dates:

Document	Version No	Submission Date
Protocol	<i>1</i>	<i>20 February 2025</i>
Information sheet	<i>1</i>	<i>20 February 2025.</i>
Consent form	<i>1</i>	<i>20 February 2025</i>
Study Instrument	<i>1</i>	<i>20 February 2025.</i>

Thank you for submitting the above documents to the KCP Research Ethics Committee (REC) for review. After a thorough evaluation, the committee has determined that your research poses minimal risk.

Please be advised that if any major changes are made to your approved research proposal, it is your responsibility to promptly notify the REC in writing. Major changes include, but are not limited to, modifications to the following:

- Research title
- Methodology
- Target population
- Conceptual framework
- Problem statement(s)
- Any other significant deviations from the original proposal

This document serves as confirmation of your eligibility to proceed to the next stage of your research.

We wish you success in your research work and look forward to supporting you in future stages of the research process.

More grace,

A handwritten signature in black ink, appearing to read 'Zaldivar E. Acodesin', written over a faint circular stamp.

Zaldivar E. Acodesin

Chairperson

Request Letter



MARCH 10, 2025

HON. ROMEO K. SALDA
Municipal Mayor
La Trinidad, Benguet



Sir,

Greetings of peace and good health!

We, the Bachelor of Science in Hospitality Management students of King's College of the Philippines are currently conducting a research study titled "Food Waste Management Strategies of Restaurants in Central Business District of La Trinidad, Benguet". In this regard, may we request of the Lists of registered Restaurants in La Trinidad for year 2024-2025 to establish our research study.

This study is only possible with the help and support of your good office. Rest assured that all information gathered shall be for academic purpose only and shall be treated with outmost confidentiality.

Rest assured that all of your information will be kept confidential under the Republic Act No. 10173, otherwise known as the Data Privacy Act.

Your favorable response to this matter will be very much appreciated. God bless and more power!

Respectfully yours,

ROSE ANN P. BINWIHAN
MARIELLE T. KILLIP
BENJIE FORD B. MOCATY
MARJON S. QUILLADAS
RHEALEE T. TAY-EW
JEYZELIE J. TERRADO

Noted:

DOLSEY WRETH P. SAPI
Instructor

Permit to Conduct Study



Republic of the Philippines
 Province of Benguet
MUNICIPALITY OF LA TRINIDAD
OFFICE OF THE MUNICIPAL MAYOR
 La Trinidad Municipal Hall, Km. 5, La Trinidad, Benguet 2601
 Tel. No.: (074) 422-2601 / 422-5618 | Fax No.: (074) 422-5621
 Official Website: www.la Trinidad.gov.ph | Email Address: mayorsit@yahoo.com

P E R M I T

NO. 92-2025

(CONDUCT STUDY/SURVEY)

PERMIT IS HEREBY GRANTED to the following, **3RD YEAR BACHELOR OF SCIENCE IN HOSPITALITY MANAGEMENT**, students from the King's College of the Philippines to conduct data gathering activities in connection with their study on "Food Waste Management Strategies of Restaurants in Central Business District of La Trinidad, Benguet" in partial fulfillment of their subject Research in Hospitality:

- a. ROSE ANN P. BINWIHAN
- b. MARIELLE T. KILLIP
- c. BENJIE FORD B. MOCATI
- d. MARJON S. QUILLADAS
- e. RHEALEE T. TAY-EW
- f. JEYZELIE J. TERRADO

They may distribute questionnaires, conduct interviews and gather data from individuals, barangays, establishments and offices in La Trinidad, Benguet provided these shall be used for the above stated purpose only. Information gathered shall be considered confidential. A copy of the study/research shall be provided the municipality through the Office of the Municipal Mayor.

They are further directed to observe minimum public health standards while conducting their activity.

Issued this 10th day of March, 2025 at La Trinidad, Benguet Province.

FOR : ROMEO K. SALDA
Municipal Mayor

BY : *[Signature]*
ATTY. ROMA C. CAYAT
Executive Assistant II

NOT VALID WITHOUT OFFICIAL SEAL

O.R #	:	9461429
Paid on	:	March 10, 2025
Paid at	:	La Trinidad, Benguet
Amount Paid	:	Php 50.00

Interview Questionnaire/ Guide Questions

Name of the restaurants(optional):

1. What strategies or practices does your restaurant currently use to manage or reduce food waste?
2. What happens to leftover food at the end of the day?
3. Do you segregate food waste? If yes, How?
4. How do you implement these food waste strategies in your restaurant?
5. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
6. What challenges do you face in implementing effective food waste management strategies?
7. How do you customer behavior and preference impact your food waste?
8. Do you provide training to your staff about food waste prevention and proper handling? How?
9. What kind of support or resources would help you improve food waste management

SAMPLE THEMATIC ANALYSIS

Coding

Name of Establishment

Dap-ayan Restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out (FIFO)

Composting

Portion Control

Repurposing Food Scraps

2. What happen to leftover food at the end of the day?

Some ingredients can be chilled or kept for other day , others are just being disposed they check the quality of food and of course they limit the food they cook and if ever there are excess food, they pass it to other branch that needs it.

3. Do you segregate food waste? If yes how?

Every trash bin has label or tags. Every time staffs do dish out, they separate the trashes immediately and accordingly after the day, they check the trash bins and fix them well, staffs and customers just know how to segregate waste.

4. How do you implement these food waste strategies in your restaurant?

-staff are using the older stock before using the newer deliveries.

- we train our kitchen and front-of-house staff on what can and can't be composted. We've set up clearly labeled bins in prep areas and dishwashing stations to make it easy.

-limiting the food being cook in the end of the day.

-customers often tend to make suggestions or tell what they want their food to be as long as it can be done by the kitchen.

-using of the peels of vegetables and bones to become stock.

-left overs like rice can recook for another dish for employee and customer.

-They pass it to another branch.

5. Do you provide training to your staff about food waste prevention and proper handling? How? yes, by providing trainings for serving seminars, orienting all staffs and OJTS

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

"Older stock is used first through proper inventory rotation to prevent spoilage."

"Surplus food or ingredients are sent to our other branch if they can use them before expiration."

7. What are challenges do you face in implementing effective food waste management strategies?

- During busy hours, it's easy for staff to grab whatever is most accessible, which sometimes leads to newer stock being used first and older items going to waste.

- some staff may forget or rush, leading to cross-contamination of waste. It takes ongoing reminders, monitoring, and refresher training to keep the system working properly.

- last-minute changes can lead to unused ingredients or extra prep, which sometimes results in more waste if not carefully managed

-not all scraps are reusable because something in busy hours we cannot collect them.

-staff are forgetting about the left-over of the rice

8. What kind of support or resources would help your improve food waste management?

We use Color-coded bins for easier waste segregation

-no we are not using bench marking data because we don't have enough time to track and compare food waste properly, our focus has been on daily operations of restaurant even on busy days.

-we use food preservation to keep our food fresh for longer and reduce waste this includes things like refrigerator, freezer, vacuum, proper packaging, and temperature control systems.

-we have staff training programs to help reduce food waste we teach our team on how to store proper food, to handle ingredients safety, and use the right portions when preparing meals.

-we choose this because the supplies consistent to give us deliver fresh and good quality of food of their product this helps us to avoid getting spoiled food, so we don't have to throw it out that helps to minimize waste to have waste reduction.

9. How do you customer behavior and preferences impact your food waste?

Some customers cannot finish their meals because of a lot of serving.

Name of Establishment

Inglay Ristorante

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out (FIFO)

Waste Segregation

Take out

2. What happen to leftover food at the end of the day?

At our restaurant, we prioritize food safety and quality above all. To ensure that all ingredients and prepared foods are safely preserved for future use, we strictly require that they are properly stored in clean, sanitized, and clearly labeled containers. Immediately after preparation or use, all items must be refrigerated without delay to maintain freshness and prevent any risk of contamination

3. Do you segregate food waste? If yes how?

Our trash bins are designed with clear labels and distinct colors, making it easy for anyone to quickly identify where to dispose of different types of waste. Each bin is strategically placed and marked according to the type of trash—be it recyclables, compostable food waste, or general refuse—ensuring that our waste management process is both efficient and environmentally responsible. This system helps prevent contamination between waste streams and supports our commitment to proper recycling and disposal practices.

4. How do you implement these food waste strategies in your restaurant?

-we cook first that was put first in the freezer.

-we throw the liquid waste at the sink like ketchup, sauces, soup while the solid waste are thrown at the specific garbage can.

- the leftovers will be taken out by the guests.

5. Do you provide training to your staff about food waste prevention and proper handling? How? yes, we train the staff by instructing them that if there are any leftovers, they will be going to ask the guest to take home their leftovers.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste? all ingredients are labeled with the delivery date as soon as they arrive.

7. What are challenges do you face in implementing effective food waste management strategies?

Some times staff mixed mix the old stock to new stock.

We lack the space in the kitchen for multiple separate bins, which makes waste sorting difficult.

Some customer preferences for oversized portions increase take-out waste

8. What kind of support or resources would help your improve food waste management?

Proper separate bins with a proper label.

9. How do you customer behavior and preferences impact your food waste?

Some customers cannot finish the food they are ordered and they ordered a lot of dishes

Name of Establishment

Ghasat Resto

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out (FIFO)

Composting

2. What happen to leftover food at the end of the day?

Kapag may natirang pagkain na pwede pang kainin, agad namin itong pinalalamig at iniimbak nang maayos sa canister para magamit sa susunod na araw."

3. Do you segregate food waste? If yes how?

We ensure a strict separation of organic and non-organic waste to promote effective waste management and sustainability. All organic waste, such as bones, vegetables, and rice, is carefully sorted and placed in designated bins, with each type of organic material kept separate.

4. How do you implement these food waste strategies in your restaurant?
We implement food waste strategies by making sure that we use older stock before the new ones.
-we reminded our staff to separate the organic waste to avoid mixing to non-organic waste
5. Do you provide training to your staff about food waste prevention and proper handling? How?
yes, by orienting them and guiding them in proper handling by demonstrating them.
-we train them by separating organic waste and non-organic waste
6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
we used first old stock before using the new stock .
we put older stock and new stock in the proper storage to be organized properly.
7. What are challenges do you face in implementing effective food waste management strategies?
Products often get pushed to the back of the fridge and forgotten.
Fruit flies and pests gather around the compost bins if they're not emptied quickly.
8. What kind of support or resources would help you improve food waste management?
Wider space for trash bins
They want a designated compost area that can dispose organic waste properly
9. How do you customer behavior and preferences impact your food waste?
Some customers food was not eaten properly
Don't eat the garnish because they're not used to taste the garnish flavor

Name of Establishment

Ali Emes Restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out (FIFO)

Portion Control

Repurposing Food Scraps

Inventory Management

2. What happen to leftover food at the end of the day?
As part of our daily routine, any leftover food that is still fresh and of good quality is immediately cooled and stored in a designated container to preserve it and make it available for employees to consume the following day. This practice ensures that diligent employees can benefit from the meals they helped prepare
3. Do you segregate food waste? If yes how?
Our staff is dedicated to maintaining an efficient and responsible waste management system by carefully segregating waste into distinct categories, such as biodegradable and non-biodegradable materials. Each team member is trained to identify and properly sort items, ensuring that biodegradable waste, such as food scraps, paper, and organic materials, are placed in dedicated bins.
4. How do you implement these food waste strategies in your restaurant?
using the stocks before new one
-we are maintain the portion control to minimize waste
- we use vegetable trimmings to make flavorful stocks and broths, and we repurpose citrus peels for house-made syrups and garnishes.
- making sure to every ingredients have to monitor
5. Do you provide training to your staff about food waste prevention and proper handling? How?
yes, by orienting them to know what will they use of the stock
6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
ingredients are stored according to FIFO (First In, First Out) to ensure older items are used first.

7. What are challenges do you face in implementing effective food waste management strategies?

Staff sometimes grab the easiest-to-reach items, not the oldest stock

Servers sometimes give larger portions to avoid complaints or to please regulars

During peak hours, there's no time to think about saving trimmings or peels

Unexpected delivery delays throw off our inventory plans

8. What kind of support or resources would help your improve food waste management?

having bins that are clearly labeled with words or pictures helps staff know exactly where to put food waste, recyclables, and general trash, which makes waste separation easier and more accurate
- bench marking data provides us with valuable insight into our operational efficiency. Comparing to other in the industry.

- We rely on advanced storage solution like vacuum packaging and controlled atmosphere storage to slow down spoilage and maintain the texture and taste of ingredient.

- A kitchen and service staff are trained in food hygiene, customer service and safety protocols.

- we choose reliable and consistent suppliers to Help us ensure that we receive high quality ingredients on time, minimizing last minute substitution and maintain menu consistency

9. How do you customer behavior and preferences impact your food waste?

Un finished food of customers

-Un eaten garnish.

-Scraps and bones

Name of Establishment

D'Exodus Gastropub

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out (FIFO)

Proper Storage

Waste Segregation

2. What happen to leftover food at the end of the day?

We implement a practice of repurposing certain leftover ingredients to create new, exciting dishes, always ensuring that food safety and quality are never compromised.

3. Do you segregate food waste? If yes how?

Staffs segregate properly by following the different trash bins for each categories

4. How do you implement these food waste strategies in your restaurant?

-the first one that is in the freezer will be the first one to be cook.

- the meat are properly stored in the refrigerator to avoid spoilage.

- ensure all staff are trained in proper waste segregation practices. We hold regular briefings and have signage above each bin to minimize mistakes.

5. Do you provide training to your staff about food waste prevention and proper handling? How? yes by regular supervision and reminders during shifts.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste? introspecting or checking the quality of freshness.

-Labeling with expirations.

7. What are challenges do you face in implementing effective food waste management strategies?

Products often get pushed to the back of the fridge and forgotten

Staff being forgotten sometimes that whys there meat left that old stock

Many employees join without knowing how to separate waste properl

8. What kind of support or resources would help your improve food waste management?

Better kitchen equipment like sealed containers to keep food fresh longer

9. How do you customer behavior and preferences impact your food waste?

-They mix their garbage

-They dont like the taste of the garnish.

-Maraming d nauubos na pagkain

Name of Establishment

`Dampa Expres

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

Portion Control

Take out

Composting

2. What happen to leftover food at the end of the day?

We have leftover food that is still fresh, safe, and of high quality, we allow our staff to take it home.

3. Do you segregate food waste? If yes how?

We use a different bins

4. How do you implement these food waste strategies in your restaurant?

We limit the portion of the food to avoid leftovers.

- I've made it a priority to reduce takeout-related waste by switching to eco-friendly, compostable packaging and giving customers the option to decline utensils, napkins, and condiments they don't need.

- we train our kitchen and staff on what can and can't be composted.

5. Do you provide training to your staff about food waste prevention and proper handling? How? Yes, during the shift of the staff the me as the manager remind them about the proper waste segregation.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste? Suppliers are chosen based on consistency and reliability, reducing the chance of poor-quality stock

7. What are challenges do you face in implementing effective food waste management strategies?

When we reduce portion sizes to cut waste, customers often feel they're not getting enough value, leading to dissatisfaction

8. What kind of support or resources would help your improve food waste management?

Training on portion control for both kitchen staff and servers.

9. How do you customer behavior and preferences impact your food waste?

The food is not eaten properly .

-They don't know the proper way to eat the seafood

Name of the establishment

Fork N'Dagger

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First Out (FIFO)

Repurposing Scrap

2. What happen to leftover food at the end of the day?

We make sure that all food is appropriately chilled, stored in clean containers, and handled with the same care we apply to all food served to our guests

3. Do you segregate food waste? If yes how?

We separate shell of seafood in another bins

By separating the waste biodegradable and non-biodegradable

4. How do you implement these food waste strategies in your restaurant?

we make sure that we used older stock before using other new stock.

- we put food scraps in a specific bin, then send them to be turned into compost

5. Do you provide training to your staff about food waste prevention and proper handling? How?

- yes, we remind our staff by sorting food waste properly.

-by continuously guiding them

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

we used first the older ingredients.

- We accurately organized the new ingredients into the proper place.

7. What are challenges do you face in implementing effective food waste management strategies?

-Products often get pushed to the back of the fridge and forgotten.

-Some are too damaged or unsafe to reuse.

8. What kind of support or resources would help your improve food waste management?

More space for trash bins.

-Wider space for storing ingredients.

9. How do you customer behavior and preferences impact your food waste?

Not all customers were take out their left overs

Name of Establishment

Taste and Cups by Joana C

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First Out (FIFO)

Portion Control

Composting

2. What happen to leftover food at the end of the day?

Any leftover food that meets our strict safety and quality standards is properly chilled and safely stored for use as staff meals the following day.

We have color coding in our trash bins.

Our kitchen supervisors ensure that our staff is separating and segregating our trash properly to ensure proper disposal

3. Do you segregate food waste? If yes how?

we do first in first out system to reduced food waste and monitoring date production and expiration.

4. How do you implement these food waste strategies in your restaurant?

- we use standardized recipes to ensure consistent portion sizes.

-we clearly labeled trash bins that our staff separating and segregating our trash properly.

5. Do you provide training to your staff about food waste prevention and proper handling? How?

by demonstrating our staff to manage the composting system. utilize ingredients across multiple dishes whenever.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

Yes, by "use first" areas were we discuss were the designating shelves for items nearly expiry

7. What are challenges do you face in implementing effective food waste management strategies?

Older stock stored at the back becomes inaccessible due to newer deliveries being stacked in front some customer expect large servings. Some staff may ignore labels due to habit or rushing during peak hours

8. What kind of support or resources would help your improve food waste management?

bench marking data is one of the resources that helps us improve food waste management. This helps us to understand customer demand and peak hours, which helps us plan how much food to prepare reducing overproduction and waste.

- another is food preservation technology we use this by refrigerating and vacuum- sealing to extend the shelf life of the ingredients. This helps us store excess ingredients properly and reduce spoilage.

- we also conduct regular staff training on portion control, proper storage and food handling practices to reduce waste during preparation and service.

- we work with reliable suppliers who deliver fresh and high-quality ingredients regularly. This helps reduces the chance of receiving spoiled or low-quality items that could go to waste.

9. How do you customer behavior and preferences impact your food waste?

Maraming mga buto ang naiiwan.

minsan kapag maraming customer yong staff ai hinalohalo yong maga basura

Name of Establishment
Mountain Chef

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First Out (FIFO)

Waste Segregation

Inventory management

2. What happen to leftover food at the end of the day?

We make it a priority to store all leftover food in clean, sanitized containers to maintain its safety and integrity.

3. Do you segregate food waste? If yes how?

We let our staff follow strict rules for waste segregation.

4. How do you implement these food waste strategies in your restaurant?

we use fifo to ensure older ingredients are used.

- we separate food waste into categories.

- We assign specific staff to check stock levels daily, and we reward accuracy to encourage consistency.

5. Do you provide training to your staff about food waste prevention and proper handling? How?

yes, we provide regular training with practical demonstration to ensure staff understand proper food handling.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste? using FIFO to ensure older item are use before newer ones expires.

7. What are challenges do you face in implementing effective food waste management strategies?

sometimes the our staff because of pressure they just grab what they see especially in peak hours.

-the customers mix their garbage

-sometimes we don't have enough time for inventory.

8. What kind of support or resources would help your improve food waste management?

Our restaurants need a ventilation of proper labeled bins

9. How do you customer behavior and preferences impact your food waste?

Customer sometimes order more than but they can't finish all the food

Name of Establishment

Health 101 Restoerant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

Composting

First in First out (FIFO)

Portion Control

Inventory Management

Waste segregation

2. What happen to leftover food at the end of the day?

We let our staff that they ensure food is not kept at room temperature longer than necessary

We use color-coded bins for different types of waste. Also reducing, reusing, and recycling food waste

3. Do you segregate food waste? If yes how?

to ensure that older stock is used before newer item, to reducing spoilage

- We measure each serving carefully to avoid food waste and keep portions consistent.

- reduce food spoilage and waste procedure for compostable from non- compostable.

- all storage areas, so ingredients are used in the order they arrive, minimizing waste from expired products."

4. How do you implement these food waste strategies in your restaurant?

We installed a three-bin system—one for compost, one for recyclables, and one for general waste—to make segregation easy and effective

5. Do you provide training to your staff about food waste prevention and proper handling? How? yes, training sessions on food waste, hygiene, waste deduction. New employees are oriented during onboarding monthly refreshers practices

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste? labeled with dates and organized so that older stock is used first

7. What are challenges do you face in implementing effective food waste management strategies? The expiry dates are difficult to read.

-Even with measuring tools, mistakes can happen, especially during peak service hours or under stress.

-The storage or composting options are not enough.

-Sometimes storage cannot properly management because. Of some deliveries not fixed it

-sometimes when peak season cannot maintain the properly storing of bins

8. What kind of support or resources would help your improve food waste management? Properly labelled bins.

-Wider space for storing ingredients.

-we use benchmark marketing to analyze customer preferences and peak dining times. This helps us prepare the right quality of food, minimizing waste caused by overproduction.

- other support that we are using is preservation technology we use this by utilizing the refrigeration and proper storage container to maintain the freshness of organic and perishable ingredients which are essential for our health- focused menu.

- we provide training to our staff on portion control, first in – First out, stock rotation and best practices in food handling to reduce preparation and storage waste.

-We also partner with reliable suppliers who understand our need for fresh, organic ingredients. This helps reduce waste due to spoilage and ensure we always receive quality products

9. How do you customer behavior and preferences impact your food waste?

They dont want the taste of the garnish.

-Maraming order pero d kaya ubusin.

Name of Establishment
Lovelifé Café and Resto

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out(FIFO)

Portion Control

Waste Segregation

Inventory Management

2. What happen to leftover food at the end of the day?

We use sometimes to prepare meals for staff

3. Do you segregate food waste? If yes how?

Segregation in place in our kitchen food waste is divided into three categories organic waste, recyclable and non-recyclable waste

4. How do you implement these food waste strategies in your restaurant?

we redesigned our storage layout so that older ingredients are always more accessible than new ones

- Offering different portion sizes on our menu—like regular and small—gives customers more control and reduces waste

- labeled bins for food waste, recyclable

- We've set par levels for every key ingredient. This helps staff reorder only when needed, preventing overstocking and keeping our storage lean and efficient.

5. Do you provide training to your staff about food waste prevention and proper handling? How? Yes, we implement our employee are **trained** by demonstrating on the importance of waste segregations in how they do the proper disposal.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste? checking deliveries for quality and freshness before acceptance their deliveries.

7. What are challenges do you face in implementing effective food waste management strategies? We sometimes mix the new stock to the old ones.

-Minsan d nasusunod ang saktong sukat lalo kong maraming guest .

-The customers mix their garbage.

Sometimes we forget to do inventory

-Sometimes our staff forgetting about the inventory because of sometimes like weeks end marami customer

8. What kind of support or resources would help your improve food waste management?

Wider space for trash bins

9. How do you customer behavior and preferences impact your food waste?

Some part of the dish are not eaten

Name of Establishment

7th Floor

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out (FIFO)

Portion Control

2. What happen to leftover food at the end of the day?

By chilling to be eat by the staff

We give the leftover food to the employee who has a pet

3. Do you segregate food waste? If yes how?

Segregate it in proper bins to be organize

4. How do you implement these food waste strategies in your restaurant?

I've trained the kitchen team to always move older items to the front and newer ones to the back when restocking shelves or fridges

- We include portion sizes in every prep and plating guide so that even new staff can follow them accurately.

5. Do you provide training to your staff about food waste prevention and proper handling? How?

Yes ,by letting our team have training.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

The team is trained to rotate stock every day and alert me if anything is nearing expiration.

7. What are challenges do you face in implementing effective food waste management strategies?

New employees may not be trained yet or may need reminders, leading to mistakes in stock rotation.

-The weighing scale is small to accommodate all the ingredients resulting waste of time.

8. What kind of support or resources would help your improve food waste management?

Wider space for storage

The bins are properly labelled.

- we use bench marking data helps us compare our performance against industry standard competitors

- we use refrigeration to extend the shelf life of perishable item like meat, dairy and vegetable.

Reduce food spoilage, maintain product quality and ensure food safety.

- we training is critical to ensuring our team follows food safety protocols, adhere to proper hygiene standard, and understand waste-reduction practices
- we also supplier who deliver fresh and high-quality ingredients on time

9. How do you customer behavior and preferences impact your food waste?

-Some of the customers dont want the take home of the food

Name of Establishment

Milo Bestro

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First Out(FIFO)

Inventory Management

2. What happen to leftover food at the end of the day?

We use leftovers creatively in staff meals when safe

Staff meals are prepared from safe, properly stored leftovers

3. Do you segregate food waste? If yes how?

waste bins are clearly labeled in the kitchen

4. How do you implement these food waste strategies in your restaurant?

make sure older ingredients are use before newer ones helping prevent spoilage.

- conduct routine kitchen checks, assign waste monitoring duties to specific staff.

5. Do you provide training to your staff about food waste prevention and proper handling? How? yes, we provide regular training to our staff. New employee oriented that include food waste

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

We maintain detailed logs for each delivery, which helps us detect and reduce recurring waste sources

7. What are challenges do you face in implementing effective food waste management strategies?

Mixing of old and new stock

8. What kind of support or resources would help your improve food waste management?

Staff can still make mistakes in rotating stock, especially if they're new or rushing during busy times.

Better communication tools to coordinate between kitchen and service staff about leftover food.

- we use bench marking data helps us compare our performance against industry standard competitors

- we use refrigeration to extend the shelf life of perishable item like meat, dairy and vegetable. Reduce food spoilage, maintain product quality and ensure food safety.

- we training is critical to ensuring our team follows food safety protocols, adhere to proper hygiene standard, and understand waste-reduction practices

- we also supplier who deliver fresh and high-quality ingredients on time.

9. How do you customer behavior and preferences impact your food waste?

So customer don't prefer take outs

Name of Establishment

Baguio Café and Restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

irst in First out (FIFO)

Take out

Waste segregation

2. What happen to leftover food at the end of the day?

By ensuring the excess food are stored in the refrigerator

3. Do you segregate food waste? If yes how?
Staff are responsible for **segregating** food waste during cleanup
4. How do you implement these food waste strategies in your restaurant?
to maintain freshness, they use the first that is put in the freezer.
-by advising the customer to take home their leftover
- to minimize the risk of contamination segregation is important by using different types of bins.
5. Do you provide training to your staff about food waste prevention and proper handling? How?
yes, they are well **trained** and oriented by checking regularly when it comes to segregation food waste and other types of waste.
6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
to **maintain** freshness, we store the delivered ingredients in the freezer
7. What are challenges do you face in implementing effective food waste management strategies?
New stock arriving often can get placed in front by mistake.
-Many customers don't want to be served a meal they can't finish and not everyone take out it.
-Without posters or pictures showing what goes in each bin, mistakes are more likely.
8. What kind of support or resources would help your improve food waste management?
he trash bins must have proper **labels**.
-Color-coded bins with simple instructions or visual guides can help reduce mistakes in waste segregation, especially for new or busy staff
- we use this because benchmarking data helps us to monitor and compare our food waste 11 levels against industry standards and this helps us identify where er can reduce waste and to improve our sustainability practices in food waste management.
-we use this resources preservation technology, we apply food precaution to prolong shelf life of our product and reduce spoilage. This will help to minimize waste and to ensure food quality for longer.
-we conduct regular training for our staff or proper handling, waste techniques reduction and proper storage to educate and aware if staff to prevent avoidable food waste.
-we work with dependable suppliers who is consistently on delivering fresh and high-quality ingredients. This reduces the risk of spoilage upon delivery and support our effort to minimize waste right from the start
9. How do you customer behavior and preferences impact your food waste?
So many leftover.

Name of Establishment
Kabunian Café & Restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?
First in First out(FIFO)
Composting
Waste Segregation
2. What happen to leftover food at the end of the day?
By preserving the food in the fridge for the next day
Sometime it consumed by our staff
Someone collects the leftover food once it accumulates
3. Do you segregate food waste? If yes how?
By **separating** compostable and non-compostable waste
Some staff nd collector they get the compostable waste to use as fertilizer foe their plants
4. How do you implement these food waste strategies in your restaurant?
we used older food to be used first and newer food supplies is stored in the cabinet.
- we implement these strategies in our restaurant by training the staff separate the compostable and non- compostable waste to avoid mixing.

- checking bins regularly to make sure waste is segregated properly.

5. Do you provide training to your staff about food waste prevention and proper handling? How?
es to make sure by giving them consistent ongoing monitoring in food waste.

- by giving them guidance and reminders.

- we **train** them to separate compostable items.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
we used **first older supplies before the new ones.**

- **putting away the expired supplies in the proper bins.**

- **we organized older supplies and the new supplies in the proper storage**

7. What are challenges do you face in implementing effective food waste management strategies?

Limited storage space for stock or deliveries.

-**sometimes when peak season we forget the properly segregating of waste**

-**They cant control the bins get full easily.**

8. What kind of support or resources would help your improve food waste management?

Availability of compostable disposal and non- compostable disposal.

-**Wider space** storage area

9. How do you customer behavior and preferences impact your food waste?

If they don't like the taste they leave their food unfinished.

-Some customers don't like **garnish they just leave it uneaten**

Name of Establishment

Nonno Guido's Ristorante

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out(FIFO)

Composting

Inventory Management

2. What happen to leftover food at the end of the day?

We only cooked food based on customer orders, so there is usually little no waste if there is any excess from what the cook has prepared, if it untouched from kitchen and staff we reheat and serve it for our family mea

3. Do you segregate food waste? If yes how?

Staffs are trained to **separate** food waste, we provide clearly labeled bins for staff

It has helped us reduce mixed waste, making disposal and composting earlier

4. How do you implement these food waste strategies in your restaurant?

We follow FIFO method to use older ingredients first.

We do some composting for organic waste.

- I regularly coordinate with chefs to plan specials around surplus or short-dated ingredients so that nothing goes to waste.

5. Do you provide training to your staff about food waste prevention and proper handling? How?
Regularly giving them reminders

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

I regularly checking inventory turnover reports to fine-tune purchasing decisions.

7. What are challenges do you face in implementing effective food waste management strategies?

Hindi clear ang label ng stock.

-Putting or sometimes we cant avoid to mix compostable waste together with trash or recyclables ruins the compost.

-Without regular checks, expired items might stay unnoticed.

8. What kind of support or resources would help you improve food waste management?

More space for composting

we use benchmarking data to compare our food cost percentage, labor efficiency and customer satisfaction rating to industry.

-We use blast chillers, walk-in coolers to store ingredients properly. This reduces spoilage, maintain food safety and helps manage inventory

-we trained staff work efficiently follow safety protocols better and customer services training also boosts employee's morale.

-choosing reliable and consistent suppliers Helps maintain food safety standard and avoid disruption in service

9. How do you customer behavior and preferences impact your food waste?

-Mixing the garbage.

-Different age groups prefer different portion sizes, affecting leftovers.

Name of Establishment

Marosan Restauran

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First Out (FIFO)

Composting

Take-out

2. What happen to leftover food at the end of the day?

We are chilling leftover foods for the next day and eat by our staff

3. Do you segregate food waste? If yes how?

We minimize contamination by properly sorting all food waste.

4. How do you implement these food waste strategies in your restaurant?

by regularly checking rotating stock.

- During closing shifts, we do a compost bin check to ensure the right items are being disposed of correctly

-by allowing our customer to take out their food

5. Do you provide training to your staff about food waste prevention and proper handling? How? yes by properly guiding them like proper disposing of waste

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

All deliveries are inspected and check by designated staff to maintain accountability and reduce errors

7. What are challenges do you face in implementing effective food waste management strategies?

Nadadaganan ang mga dating naka lagay sa freezer.

it produces odor especially the vegetable peels if it is made as compost we cant avoid its odor

-Some customers dont want to take home their leftovers.

8. What kind of support or resources would help you improve food waste management?

Separate trash bins with proper labelling

we use benchmarking data to assess which menu item are most popular and when customer volume is high. This allows us to adjust our food preparation accordingly and avoid excess.

- We are also using this. By utilizing the refrigeration and proper containers to maintain the freshness of organic and perishable ingredients which are essential for our health-focused menu.

- We provide training to our staff on portion control, first in – first out, stock rotation, and best practices in food handling to reduce preparation and storage waste.

- we partner with reliable suppliers who understand our need for fresh, organic ingredients. This help reduce waste due to spoilage and ensure we always receive quality products

9. How do you customer behavior and preferences impact your food waste?
 They complain on the taste if the food.
 -They have so many leftover

Name of Establishment
 Kalie's Grill and Restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?
 First in, First Out(FIFO)
 Portion Control

2. What happen to leftover food at the end of the day?
 Will be put in the freezer then in the morning it will be the first to be served, the vegetable that been cooked can be consumed by the staff

3. Do you segregate food waste? If yes how?
 Vegetable scraps are separated for composting

4. How do you implement these food waste strategies in your restaurant?
 Rotate our stock to ensure older item are used first.
 Prevent over serving and encourage customer to take leftover.

5. Do you provide training to your staff about food waste prevention and proper handling? How?
 yes, monthly training session that cover waste. New staff are trained during onboarding as well.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
 All items are stored with clear labels, including dates and usage instructions.

7. What are challenges do you face in implementing effective food waste management strategies?
 Doesn't clearly show expiration dates.
 -Some customers have different expectations in portions size of the food

8. What kind of support or resources would help your improve food waste management?
 Wider space for ingredients.
 -Proper inventory
 -we use benchmarking data to see how much food we waste compared to other, by this, it helps us to find ways to waste less.
 -we use food preservation methods to keep for more freshness for a longer time. This helps us to avoid throwing food away because it goes bad too quickly.
 -we train our staff on how to handle food properly, store it correctly, and avoid waste. When staff know what to do to lessen the food that get wasted.
 - we choose suppliers who always give us good quality delivery of food and who deliver fresh food, this will help us to avoid spoiled products

9. How do you customer behavior and preferences impact your food waste?
 They cant eat their orders.
 -They dont know to segregate their garbage they just mix it.

Name of Establishment
 Al's Restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?
 First in First Out (FIFO)
 Composting
 Waste segregation

2. What happen to leftover food at the end of the day?
 Staff meals are prepared from safe, properly stored leftovers

3. Do you segregate food waste? If yes how?

We **separate** food waste from regular trash every day

4. How do you implement these food waste strategies in your restaurant?

To avoid spoilage of the food and to maintain the freshness of the food they cook first the first that they purchase before the latest.

- Some food items are composted to recycle nutrients back into the soil.

- We provides waste bins for biodegradable and non- biodegradable and staff are responsible in maintaining the trash bins

5. Do you provide training to your staff about food waste prevention and proper handling? How? we **instruct** the staff about the proper waste prevention lalo na sa segregation.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste? **Checking all the deliveries** of all the ingredient

7. What are challenges do you face in implementing effective food waste management strategies? **We cany avoid mixing the new stock to the old stock.**

8. What kind of support or resources would help your improve food waste management?

- **Not enough space for composting.**

- **The customers mix their garbage kaya nahihirapan kame i segregate ang mga basura.** **More space** for sorting and storing waste properly.

- **we don't use bench marking data because we don't have system to measure how much food we waste. Right now, we just manage it by observing but it's not very accurate.**

- **we use food preservation technology to extend the shelf life of the products, to maintain safety food and to reduce waste. It ensures our customer receive high quality and fresher items.**

- **we train them by teaching them how to separate food that that can still be used, donated or composted. By this we throw away much less waste.**

- **Because good suppliers deliver on time in a good condition this helps us plan better and avoid waste from delays so spoiled items.**

9. How do you customer behavior and preferences impact your food waste? Customers reject certain sides or **garnishes, leading to waste.**

Name of Establishment

Rebecas Restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?

First in First out (FIFO)

Waste Segregation

Composting

2. What happen to leftover food at the end of the day?

Chilling the left over foods for the next day for the staff to eat

3. Do you segregate food waste? If yes how?

3We have color coding in our trash bins.

Our kitchen supervisors ensure that our staff is **separating and segregating** our trash properly to ensure proper

4. How do you implement these food waste strategies in your restaurant?

- **we cook first the older stock to be consumed first.**

- **Not over mixing the trash we provide trash bin each.**

- **we compost the composable ones by stocking them and someone we know will collect them.**

5. Do you provide training to your staff about food waste prevention and proper handling? How? yes, by regularly **reminding** them if the trash been full, they change it immediately.

- by demonstrating them first in properly handling correctly in their first day of work in us

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
we cooked first the older stock.
7. What are challenges do you face in implementing effective food waste management strategies?
The older stock is tampered with the new ones that why sometimes instead of the older stock we use the new ones.
-With somuch trash they dont even know where to place them.
-It attracts insects.
8. What kind of support or resources would help your improve food waste management?
Wider space for trash bins to be put more.
-benchmarking data helps us compare our food cost customer satisfaction and operational efficiency against industry.
-freezing help extend the shelf life of ingredients, reduce waste, and maintain food safety.
-we are training cover food safety allergen handling and customer service. Well-trained staff help minimize mistake and reduce waste.
-Supplier deliver fresh ingredients on schedule, support consistent food quality and menu availability
9. How do you customer behavior and preferences impact your food waste?
some customers cannot fully eat their food. Some are ashamed to take left overs.

Name of Establishment
J's restaurant

1. What strategies or practices does your restaurant currently use to manage or reduce waste?
IFFirst In First Out Waste Segregation
2. What happen to leftover food at the end of the day?
Someone we know will collect them. for pigs food "arasaw" every morning
3. Do you segregate food waste? If yes how?
We have separate garbage bins for the leftover foods and non-biodegradable materials
4. How do you implement these food waste strategies in your restaurant?
We use older stocks before using newly purchased goods
5. Do you provide training to your staff about food waste prevention and proper handling? How?
Yes, staffs are oriented and trained about food waste that moment they started working in the restaurant
6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?
They are properly stored for example some vegetables are stored in chiller to prevent food waste and we used older stocks
7. What are challenges do you face in implementing effective food waste management strategies?
Mostly during busy hours the staff do not properly segregate trash.
8. What kind of support or resources would help your improve food waste management?
Having reliable suppliers made it sure our ingredients are always frwesh
9. How do you customer behavior and preferences impact your food waste?
Some of the customers don't want to take out
Customers sometimes mix their garbages

THEMES

1. What are strategies or practices does your restaurant currently use to manage or reduce waste?

	First In First Out	Composting	Portion Control,	Repurposing Food Scraps	Take Out	inventory Management	Waste segregation
Number of respondents	20	13	12	4	2	6	8

2. What happen to leftover food at the end of the day?

Chilling

Some ingredients can be chilled or kept for other day , others are just being disposed they check the quality of food and of course they limit the food they cook and if ever there are excess food, they pass it to other branch that needs it.

We make sure that all food is appropriately chilled, stored in clean containers, and handled with the same care we apply to all food served to our guests.

Any leftover food that meets our strict safety and quality standards is properly chilled and safely stored for use as staff meals the following day.

By chilling to be eat by the staff

By ensuring the excess food are stored in the refrigerator

By preserving the food in the fridge for the next day

We are chilling leftover foods for the next day and eat by our staff

Will be put in the freezer then in the morning it will be the first to be served, the vegetable that been cooked can be consumed by the staff

Chilling the left-over foods for the next day for the staff to eat

Immediately after preparation or use, all items must be refrigerated without delay to maintain freshness and prevent any risk of contamination

Stored in containers

At our restaurant, we prioritize food safety and quality above all. To ensure that all ingredients and prepared foods are safely preserved for future use, we strictly require that they are properly stored in clean, sanitized, and clearly labeled containers.

Kapag may natirang pagkain na pwede pang kainin, agad namin itong pinalalamig at iniimbak nang maayos sa canister para magamit sa susunod na araw."

As part of our daily routine, any leftover food that is still fresh and of good quality is immediately cooled and stored in a designated container to preserve it and make it available for employees to consume the following day. This practice ensures that diligent employees can benefit from the meals they helped prepare.

We make it a priority to store all leftover food in clean, sanitized containers to maintain its safety and integrity.

Repurposing

We implement a practice of repurposing certain leftover ingredients to create new, exciting dishes, always ensuring that food safety and quality are never compromised.

Leftovers are for the Staff

We only cooked food based on customer orders, so there is usually little no waste if there is any excess from what the cook has prepared, if it untouched from kitchen and staff we reheat and serve it for our family meal

We have leftover food that is still fresh, safe, and of high quality, we allow our staff to take it home.

We let our staff that they ensure food is not kept at room temperature longer than necessary

We use sometimes to prepare meals for staff

We give the leftover food to the employee who has a pet

We use leftovers creatively in staff meals when safe, Staff meals are prepared from safe, properly stored leftovers

We only cooked food based on customer orders, so there is usually little no waste if there is any excess from what the cook has prepared, if it untouched from kitchen and staff we reheat and serve it for our family meal

Sometime it consumed by our staff

Staff meals are prepared from safe, properly stored leftovers

3. Do you segregate food waste? If yes how?

LABELED BINS

Every trash bin has label or tags. Every time staffs do dish out, they separate the trashes immediately and accordingly after the day, they check the trash bins and fix them well, staffs and customers just know how to segregate waste.

Our trash bins are designed with clear labels and distinct colors, making it easy for anyone to quickly identify where to dispose of different types of waste. Each bin is strategically placed and marked according to the type of trash—be it recyclables, compostable food waste, or general refuse—ensuring that our waste management process is both efficient and environmentally responsible. This system helps prevent contamination between waste streams and supports our commitment to proper recycling and disposal practices.

waste bins are clearly labeled in the kitchen

SEGREGATION OR SEPARATION

Our staff is dedicated to maintaining an efficient and responsible waste management system by carefully segregating waste into distinct categories, such as biodegradable and non-biodegradable materials. Each team member is trained to identify and properly sort items, ensuring that

biodegradable waste, such as food scraps, paper, and organic materials, are placed in dedicated bins.

We ensure a strict separation of organic and non-organic waste to promote effective waste management and sustainability. All organic waste, such as bones, vegetables, and rice, is carefully sorted and placed in designated bins, with each type of organic material kept separate.

Staffs segregate properly by following the different trash bins for each categories

We separate shell of seafood in another bins , By separating the waste biodegradable and non-biodegradable

Our kitchen supervisors ensure that our staff is separating and segregating our trash properly to ensure proper disposal.

We let our staff follow strict rules for waste segregation.

Segregation in place in our kitchen food waste is divided into three categories organic waste, recyclable and non-recyclable waste

Segregate it in proper bins to be organize

Staff are responsible for segregating food waste during cleanup

By separating compostable and non-com

Staffs are trained to separate food waste, we provide clearly labeled bins for staff .It has helped us reduce mixed waste, making disposal and composting earlier

Vegetable scraps are separated for composting

COLOR-CODED BINS

We use color-coded bins for different types of waste. Also reducing, reusing, and recycling food waste

We use a different bins

MINIMIZE CONTAMINATION

We separate food waste from regular trash every day

Our kitchen supervisors ensure that our staff is separating and segregating our trash properly to ensure proper.

We minimize contamination by properly sorting all food waste.

4. How do you implement these food waste strategies in your restaurant?

Stock Rotation

I've trained the kitchen team to always move older items to the front and newer ones to the back when restocking shelves or fridges

staff are using the older stock before using the newer deliveries.

we cook first that was put first in the freezer.

We implement food waste strategies by making sure that we use older stock before the new ones. using the stocks before new one.

the first one that is in the freezer will be the first one to be cook.

we make sure that we used older stock before using other new stock.

we do first in first out system to reduced food waste and monitoring date production and expiration.
we use fifo to ensure older ingredients are used to ensure that older stock is used before newer item,
to reducing spoilage

we redesigned our storage layout so that older ingredients are always more accessible than new ones
make sure older ingredients are use before newer ones helping prevent spoilage.

to maintain freshness, they use the first that is put in the freezer.

we used older food to be used first and newer food supplies is stored in the cabinet.

We follow FIFO method to use older ingredients first.

by regularly checking rotating stock.

Rotate our stock to ensure older item are used first.

To avoid spoilage of the food and to maintain the freshness of the food they cook first the first that
they purchase before the latest.

we cook first the older stock to be consumed first.

Staff training and compliance

During closing shifts, we do a compost bin check to ensure the right items are being disposed of
correctly

we cook first the older stock to be consumed first.

we reminded or staff to separate the organic waste to avoid mixing to non-organic waste

we train our kitchen and staff on what can and can't be composted.

we clearly labeled trash bins that our staff separating and segregating our trash properly.

reduce food spoilage and waste procedure for compostable from non- compostable

we implement these strategies in our restaurant by training the staff separate the compostable and
non- compostable waste to avoid mixing.

We do some composting for organic waste.

During closing shifts, we do a compost bin check to ensure the right items are being disposed of
correctly Some food items are composted to recycle nutrients back into the soil.

we compost the compostable ones by stocking them and someone we know will collect them.

Portion Management

limiting the food being cook in the end of the day customers often tend to make suggestions or tell
what they want their food to be as long as it can be done by the kitchen.

we are maintain the portion control to minimize waste

We limit the portion of the food to avoid leftovers.

we use standardized recipes to ensure consistent portion sizes.

We measure each serving carefully to avoid food waste and keep portions consistent.

Offering different portion sizes on our menu—like regular and small—gives customers more
control and reduces waste

We include portion sizes in every prep and plating guide so that even new staff can follow them
accurately.

Prevent over serving and encourage customer to take leftover.

Resourcefulness

using of the peels of vegetables and bones to become stock, left overs like rice can recook for another dish for employee and customer.,They pass it to another branch.

we use vegetable trimmings to make flavorful stocks and broths, and we repurpose citrus peels for house-made syrups and garnishes.

we put food scraps in a specific bin, then send them to be turned into compost

Proper segregation practices

we throw the liquid waste at the sink like ketchup, sauces, soup while the solid waste are thrown at the specific garbage

ensure all staff are trained in proper waste segregation practices. We hold regular briefings and have signage above each bin to minimize mistakes.

we separate food waste into categories.

We installed a three-bin system—one for compost, one for recyclables, and one for general waste—to make segregation easy and effective

labeled bins for food waste, recyclable

to minimize the risk of contamination segregation is important by using different types of bins.

checking bins regularly to make sure waste is segregated properly.

We provides waste bins for biodegradable and non- biodegradable and staff are responsible in maintaining the trash bins

Not over mixing the trash we provide trash bin each.

Proper stored

I've made it a priority to reduce takeout-related waste by switching to eco-friendly, compostable packaging and giving customers the option to decline utensils, napkins, and condiments they don't need. the leftovers will be taken out by the guests. by advising the customer to take home their leftover by allowing our customer to take out their food

Monitor

we've set par levels for every key ingredient. This helps staff reorder only when needed, preventing overstocking and keeping our storage lean and efficient.

making sure to every ingredients have to monitor

We assign specific staff to check stock levels daily, and we reward accuracy to encourage consistency.

all storage areas, so ingredients are used in the order they arrive, minimizing waste from expired products."

5. Do you provide training to your staff about food waste prevention and proper handling?
How?

Trainings

yes, by providing trainings for serving seminars, orienting all staffs and OJTS

yes, we train the staff by instructing them that if there are any leftovers, they will be going to ask the guest to take home their leftovers.

we train them by separating organic waste and non-organic waste.

- yes, we provide regular training with practical demonstration to ensure staff understand proper food handling.

yes, training sessions on food waste, hygiene, waste deduction.

Yes, we implement our employee are trained by demonstrating on the importance of waste segregations in how they do the proper disposal.

Yes ,by letting our team have training.

yes, we provide regular training to our staff. New employee oriented that include food waste.

yes, they are well trained and oriented by checking regularly when it comes to segregation food waste and other types of waste.

we train them to separate compostable items.

yes, monthly training session that cover waste. New staff are trained during onboarding as well.

Orientation

yes by regular supervision and reminders during shifts.

Yes, during the shift of the staff the me as the manager remind them about the proper waste segregation.

yes, we remind our staff by sorting food waste properly.-by continuously guiding them

yes, by regularly reminding them if the trash been full, they change it immediately, by demonstrating them first in properly handling correctly in their first day of work in us.

Reminders

yes, by orienting them and guiding them in proper handling by demonstrating them.

yes, by orienting them to know what will they use of the stock.

Demonstration

Yes, by “use first” areas were we discuss were the designating shelves for items nearly expiry. by demonstrating our staff to manage the composting system.

New employees are oriented during onboarding monthly refreshers practices.

Instruction

we instruct the staff about the proper waste prevention lalo na sa segregation.

6. What steps do you take from the moment ingredients are delivered to ensure minimal waste?

Old stock and New Stock

Older stock is used first through proper inventory rotation to prevent spoilage."

we used first old stock before using the new stock , we put older stock and new stock in the proper storage to be organized properly.

we used first the older ingredients, We accurately organized the new ingredients into the proper place.

using FIFO to ensure older item are use before newer ones expires.

we used first older supplies before the new ones.

we cooked first the older stock.

we organized older supplies and the new supplies in the proper storage

Labeling with expirations

all ingredients are labeled with the delivery date as soon as they arrive.

ingredients are stored according to FIFO (First In, First Out) to ensure older items are used first.

Labeling with expirations.

The team is trained to rotate stock every day and alert me if anything is nearing expiration.

All items are stored with clear labels, including dates and usage instructions.

labeled with dates and organized so that older stock is used first

Surplus food or ingredients are sent to our other branch if they can use them before expiration."

putting away the expired supplies in the proper bins.

Choosing Suppliers

Suppliers are chosen based on consistency and reliability, reducing the chance of poor-quality stock

Checking

checking deliveries for quality and freshness before acceptance their deliveries.

Checking all the deliveries of all the ingredients

I regularly checking of inventory turnover reports to fine-tune purchasing decisions.

All deliveries are inspected and check by designated staff to maintain accountability and reduce errors

inspecting or checking the quality of freshness.

7. What are challenges do you face in implementing effective food waste management strategies?

Mixing Stock

During busy hours, it's easy for staff to grab whatever is most accessible, which sometimes leads to newer stock being used first and older items going to waste.

Some times staff mixed mix the old stock to new stock.

Products often get pushed to the back of the fridge and forgotten.

Staff sometimes grab the easiest-to-reach items, not the oldest stock

Products often get pushed to the back f the fridge and forgotten

sometimes the our staff because of pressure they just grab what they see especially in peak hours.

The expiry dates are difficult to read .

We sometimes mix the new stock to the old ones.

New employees may not be trained yet or may need reminders, leading to mistakes in stock rotation.

Mixing of old and new stock

New stock arriving often can get placed in front by mistake.

Limited storage space for stock or deliveries.

Hindi clear ang label ng stock.

Nadadaganan ang mga dating naka lagay sa freezer.

Doesn't clearly show expiration dates.

We can avoid mixing the new stock to the old stock.

The older stock is tampered with the new ones that why sometimes instead of the older stock we use the new ones.

Cross-contamination

some staff may forget or rush, leading to cross-contamination of waste. It takes ongoing reminders, monitoring, and refresher training to keep the system working properly.

Fruit flies and pests gather around the compost bins if they're not emptied quickly.

The storage or composting options are not enough.

sometimes hen peak season we forget the properly segregating of waste

Putting or sometimes we cant avoid to mix compostable waste together with trash or recyclables ruins the compost.

-It attracts insects

We lack the space in the kitchen for multiple separate bins, which makes waste sorting difficult.

Incorrect Measurements

Servers sometimes give larger portions to avoid complaints or to please regulars

When we reduce portion sizes to cut waste, customers often feel they're not getting enough value, leading to dissatisfaction

some customer expect large servings.

Even with measuring tools, mistakes can happen, especially during peak service hours or under stress.

Minsan d nasusunod ang saktong sukat lalo kong maraming guest .

The weighing scale is small to accommodate all the ingredients resulting waste of time.

Some customers have different expectations in portions size of the food .

Not enough time

Last-minute changes can lead to unused ingredients or extra prep, which sometimes results in more waste if not carefully managed

not all scraps are reusable because something in busy hours we cannot collect them., staff are forgetting about the left-over of the rice

During peak hours, there's no time to think about saving trimmings or peels

Improper waste segregation

Some are too damaged or unsafe to reuse

Some staff may ignore labels due to habit or rushing during peak hours.

It produces odor especially the vegetable peels if it is made as compost we cant avoid its odor

Not enough space for composting.

Many employees join without knowing how to separate waste properly

the customers mix their garbage

sometimes when peak season cannot maintain the properly storing of bins

The customers mix their garbage.

Without posters or pictures showing what goes in each bin, mistakes are more likely.

they can't control the bins get full easily.

The customers mix their garbage kaya nahihirapan kame i segregate ang mga basura.

With somuch trash they dont even know where to place them.

Unfinish Meal

Some customer preferences for oversized portions increase take-out waste

Many customers don't want to be served a meal they can't finish and not everyone take out it.

Some customers dont want to take home their leftovers.

Forgetting the Stock

taff being forgotten sometimes that whys there meat left that old stock

Inventory control deficiencies

Unexpected delivery delays throw off our inventory plans

sometimes we don't have enough time for inventory.

sometimes storage cannot properly management because. Of some deliveries not fixed it

Sometimes our staff forgetting about the inventory because of sometimes like weeks end marami customer

Staff can still make mistakes in rotating stock, especially if they're new or rushing during busy times.

Without regular checks, expired items might stay unnoticed.

8. What kind of support or resources would help your improve food waste management?

Colored bins

We use Color-coded bins for easier waste segregation

Color-coded bins with simple instructions or visual guides can help reduce mistakes in waste segregation, especially for new or busy staff

Separate bins

Proper separate bins with a proper label.

Separate trash bins with proper labelling
waste separation easier and more accurate

Wider space

Wider space for trash bins, They want a designated compost area that can dispose organic waste properly

Wider space for storing ingredients

Wider space for trash bins so that the street dogs and cats will not scatter the garbage

Wider space for storing ingredients.

Wider space for trash bins

Wider space storage area

More space for composting

Wider space for ingredients.

More space for sorting and storing waste properly.

Wider space for trash bins to be put more.

Labelled Bins

Having bins that are clearly labeled with words or pictures helps staff know exactly where to put food waste, recyclables, and general trash, which makes waste separation easier and more accurate
Properly labelled bins

The bins are properly labelled.

The trash bins must have proper labels.

Our restaurants need a ventilation of proper labeled bins .

Better Communication

Better communication tools to coordinate between kitchen and service staff about leftover food.

Training on portion control

Training on portion control for both kitchen staff and servers.

Sealed Container

Better kitchen equipment like sealed containers to keep food fresh longer

Bench Marking

--no we are not using bench marking data because we don't have enough time to track and compare food waste properly, our focus has been on daily operations of restaurant even on busy days.

- bench marking data provides us with valuable insight into our operational efficiency. Comparing to other in the industry

-- bench marking data is one of the resources that helps us improve food waste management. This helps us to understand customer demand and peak hours, which helps us plan how much food to prepare reducing overproduction and waste.

we use benchmark marketing to analyze customer preferences and peak dining times. This helps us prepare the right quality of food, minimizing waste caused by overproduction.

- we use bench marking data helps us compare our performance against industry standard competitors

we use this because benchmarking data helps us to monitor and compare our food waste ll levels against industry standards and this helps us identify where er can reduce waste and to improve our sustainability practices in food waste management

we use bench marking data to compare our food cost percentage, labor efficiency and customer satisfaction rating to industry.

we use bench marking data to assess which menu item are most popular and when customer volume is high. This allows us to adjust our food preparation accordingly and avoid excess.

we use benchmarking data to see how much food we waste compared to other, by this, it helps us to find ways to waste less.

we don't use bench marking data because we don't have system to measure how much food we waste. Right now, we just manage it by observing but it's not very accurate.

benchmarking data helps us compare our food cost customer satisfaction and operational efficiency against industry.

CHOOSING RELIABLE AND CONSISTENT SUPPLIER

we choose this because the supplies consistent to give us deliver fresh and good quality of food of their product this helps us to avoid getting spoiled food, so we don't have to throw it out that helps to minimize waste to have waste reduction

we work with reliable suppliers who deliver fresh and high-quality ingredients regularly. This helps reduces the chance of receiving spoiled or low-quality items that could go to waste

- we choose reliable and consistent suppliers to Help us ensure that we receive high quality ingredients on time, minimizing last minute substitution and maintain menu consistency.

We also partner with reliable suppliers who understand our need for fresh, organic ingredients. This helps reduce waste due to spoilage and ensure we always receive quality products.

- we also supplier who deliver fresh and high-quality ingredients on time.

we also choose consistent suppliers to minimize waste generation and re use materials whenever possible.

we work with dependable suppliers who is consistently on delivering fresh and high-quality ingredients. This reduces the risk of spoilage upon delivery and support our effort to minimize waste right from the start.

choosing reliable and consistent suppliers Helps maintain food safety standard and avoid disruption in service.

we partner with reliable suppliers who understand our need for fresh, organic ingredients. This help reduce waste due to spoilage and ensure we always receive quality products.

we choose suppliers who always give us good quality delivery of food and who deliver fresh food, this will help us to avoid spoiled products.

Because good suppliers deliver on time in a good condition this helps us plan better and avoid waste from delays so spoiled items.

Supplier deliver fresh ingredients on schedule, support consistent food quality and menu availability

9. How do you customer behavior and preferences impact your food waste?

Unfinished meals

Some customers cannot finish their meals because of a lot of serving.

Some customers cannot finish the food they are ordered and they ordered a lot of dishes

Un finished food of customers

Maraming d nauubos na pagkain

Maraming order pero d kaya ubusin.

So many leftover by the customers

Customer sometimes order more than but they can't finish all the food

Customer induced food waste

Some customers food was not eaten properly

The food is not eaten properly

Some part of the dish are not eaten

They cant eat their orders.

some customers cannot fully eat their food.

Un eaten garnish

Don't eat the garnish because they're not used to taste the garnish flavor

Un eaten garnish.

They dont like the taste of the garnish.

They dont want the taste of the garnish

Some customers don't like garnish they just leave it uneaten

Customers reject certain sides or garnishes, leading to waste.

Don't prefer take out

Not all customers were take out their left overs

Some of the customers dont want the take home of the food

So customer don't prefer take outs

Left bones

Maraming mga buto ang naiiwan.

Mixed waste

minsan kapag maraming customer yong staff ai hinalohalo yong maga basura

CURRICULUM VITAE

NAME: Rose ann P. Binwihan
Address: Pugo Baculongan Norte, Buguias, Benguet

PERSONAL INFORMATION

Date of Birth: October 05, 2004
Place of Birth: Atok Benguet
Citizenship: Filipino
Civil Status: Single
Provincial Address: Pugo, Baculongan Norte, Buguias, Benguet

EDUCATION

Tertiary Level: King's College of The Philippines
Pico, La Trinidad, Benguet
Secondary Level: Loo National High School
Loo, Buguias, Benguet
Elementary: Pugo James Mocate Elementary School
Pugo Baculongan Norte, Buguias, Benguet

SEMINARS ATTENDED

Seminar and Hotel Familiarization

Le Monet Hotel - Camp John Hay, Baguio
May 12-13, 2023

Food Handlers' Orientation Seminar

Kings College of the Philippines - Pico, La Trinidad-Benguet
October 20, 2022

Basic First Aid Training

Kings College of the Philippines - Pico, La Trinidad-Benguet
October 7, 2022

CURRICULUM VITAE

NAME: Marielle T. Kilip
Address: Balili La Trinidad Benguet

PERSONAL INFORMATION

Date of Birth: March 4, 2002
Place of Birth: Baculongan Norte, Buguias ,Benguet
Citizenship: Filipino
Civil Status: Single
Provincial Address: Buguias, Benguet

EDUCATION

Tertiary Level King's College of the Philippines
Pico, La Trinidad, Benguet
Secondary Level San Jose High School
Poblacion, La Trinidad, Benguet
Elementary Balili Elementary School
Balili, La Trinidad, Benguet

SEMINARS ATTENDED

Seminar and Hotel Familiarization

Le Monet Hotel - Camp John Hay, Baguio
May 12-13, 2023

Food Handlers' Orientation Seminar

Kings College of the Philippines - Pico, La Trinidad-Benguet
October 20, 2022

Basic First Aid Training

Kings College of the Philippines - Pico, La Trinidad-Benguet
October 7,2022

CURRICULUM VITAE

NAME: Benjie Ford B. Mocati
Address: Shilan, La Trinidad, Benguet

PERSONAL INFORMATION

Date of Birth: March 19, 2003
Place of Birth: Catlubong, Buguias, Benguet
Citizenship: Filipino
Civil Status: Single
Provincial Address: Amlimay, Buguias, Benguet

EDUCATION

Tertiary Level: King's College of the Philippines
Pico, La Trinidad, Benguet
Secondary Level: Cordillera Career Development College
Buyagan, La Trinidad, Benguet
Elementary: Buyagan Elementary School
Buyagan, La Trinidad, Benguet

SEMINARS ATTENDED

Seminar and Hotel Familiarization

Le Monet Hotel - Camp John Hay, Baguio
May 12-13, 2023

Food Handlers' Orientation Seminar

Kings College of the Philippines - Pico, La Trinidad-Benguet
October 20, 2022

Basic First Aid Training

Kings College of the Philippines - Pico, La Trinidad-Benguet
October 7, 2022

CURRICULUM VITAE

NAME: Marjon S. Quilladas
Address: Balili, La Trinidad, Benguet

PERSONAL INFORMATION

Date of Birth: October 19, 2004
Place of Birth: ITRMC
Citizenship: Filipino
Civil Status: Single
Provincial Address: Salangsang, Bumbuneg, San Gabriel-La Union

EDUCATION

Tertiary Level King's College of the Philippines
Pico, La Trinidad-Benguet
Secondary Level San Gabriel Vocational High School
San Gabriel-La Union
Elementary San Gabriel Central School
San Gabriel-La Union

SEMINARS ATTENDED

Filipino Brand of Service Excellence

Kings College of the Philippines - Pico, La Trinidad-Benguet
April 25, 2025

Seminar and Hotel Familiarization

Le Monet Hotel - Camp John Hay, Baguio
May 12-13, 2023

Food Handlers' Orientation Seminar

Health Services Office - Baguio City
November 22, 2022

Basic First Aid Training

Kings College of the Philippines - Pico, La Trinidad-Benguet
October 7, 2022

CURRICULUM VITAE

NAME: Rhealee T. Tay-ew
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PERSONAL INFORMATION

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EDUCATION

Tertiary Level King's College of the Philippines
Pico, La Trinidad, Benguet
Secondary Level Bakun National High School Main
Poblacion, Bakun, Benguet
Elementary Palina Elementary School
Palina , Kibungan, Benguet

SEMINARS ATTENDED

Seminar and Hotel Familiarization
Le Monet Hotel - Camp John Hay, Baguio
May 12-13, 2023

Food Handlers' Orientation Seminar
Kings College of the Philippines - Pico, La Trinidad, Benguet
October 20, 2022

Basic First Aid Training
Kings College of the Philippines - Pico, La Trinidad-Benguet
October 7, 2022

CURRICULUM VITAE

NAME: Jeyzelie J. Terrado
Address: Cot-Cot, Bangao, Buguias, Benguet

PERSONAL INFORMATION

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EDUCATION

Tertiary Level King's College of the Philippines
Pico, La Trinidad, Benguet
Secondary Level Loo National High School
Loo, Buguias, Benguet
Elementary Cot-Cot Talabis Elementary School
Cot-Cot, Bangao, Buguias, Benguet

SEMINARS ATTENDED

Seminar and Hotel Familiarization

Le Monet Hotel - Camp John Hay, Baguio
May 12-13, 2023

Food Handlers' Orientation Seminar

Kings College of the Philippines - Pico, La Trinidad, Benguet
October 20, 2022

Basic First Aid Training

Kings College of the Philippines - Pico, La Trinidad, Benguet
October 7, 2022